

Investment Sensitivity to Lender Default Shocks*

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ABSTRACT

We investigate how idiosyncratic lender shocks impact corporate investment. Lenders with recent default experience write stricter loan contracts, leading to a reduction in real investment for borrowing firms. The decline in investment is not attributable to loan riskiness, borrowers agency costs, the lender-borrower relationship nexus, or to lender capitalization, but is less pronounced when non-bank lenders increase. The findings remain robust when controlling for lender fragility, macroeconomic conditions, aggregate defaults in the economy, and excluding defaulters from the same industry or geographic region. The evidence suggests that defaults inform lenders about investment opportunities and their screening ability, and adjustments to this information have real economic consequences.

JEL classification: E22, G21, G31, G32, G33.

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I. Introduction

Managers must make investment choices under uncertainty based on their forecast of future consumer demand or future prices (Lucas and Prescott (1971)). Similarly, lenders make loan decisions and write credit contracts based on both hard and soft information known at the time of contracting. Under uncertainty, agents can learn from the past (e.g., Marcet and Sargent (1989)). Recent research has investigated how lenders learn from shocks within their lending portfolio and modify their screening and contracting processes (Murfin (2012)). In this paper, we examine how these supply-side shocks to lender screening affect the corporate investment decisions of firms.

Although the broad association between financing and investment are well established (e.g., Stein (2003), Chava and Roberts (2008)), the precise mechanisms that drive the relationship are relatively less known. A large literature focuses on the role of demand side factors, but are relatively silent about how supply-side frictions influence firm investment. Murfin (2012) examines how lender's recent default experience can influence the strictness of their loan contracts and finds that banks write stricter contracts compared to peer banks. The results suggest that banks learn about their own screening process when they experience defaults in their portfolio. We employ this idiosyncratic shock to credit terms to examine investment choices of affiliated firms. We find that this supply-side credit shock leads to a reduction in corporate investment for other firms borrowing from the bank.

We focus our attention on recent default events because large corporate defaults provide a unique opportunity for several reasons. First, corporate defaults damage lender profitability significantly and has the potential to affect lending behavior. For example, Chava and Purnanandam (2011) show that banks affected by the Russian sovereign debt default reduce lending. Gopalan, Nanda, and Yerramilli (2011) show that lender defaults dent lead arranger reputations and negatively affect their activity in the syndicated loan market. A reduction in supply side lending activity following default shocks suggests possible variation in the intensive margin of lending by affecting the terms of new lending and impacting borrowers' investment. Second, defaults may alter the beliefs and processes of the lender. Defaults contain valuable information for the lenders, allowing them to update their beliefs about economic conditions. Also, defaults can lead to lenders reassessing their own credit risk models and internal controls (Murfin (2012)), thereby influencing future lending arrangements. Third, to the extent that default risk in the lender portfolio is correlated, it is easier for the lender to make alterations incrementally through new loans rather than reassessing the entire loan portfolio. Reassessing the entire portfolio is made difficult by state contingent allocation of control rights and ex-post limits to managerial attention.

How can supply-side frictions affect borrower's investment? Ideally, if lenders make independent lend-

ing decisions, defaults in a few loans should not affect other borrowers. However, we hypothesize that a lender’s default experience may reduce borrower investment for several possible reasons. First is a ‘learning’ hypothesis where lenders learn about their own credit assessment ability and evaluation of future investment opportunities. Defaults contain valuable information about possible weaknesses in the lender’s screening and monitoring mechanisms as well as information about the investment opportunities in the economy. A lender who believes that having control rights earlier can minimize losses will prefer a transfer of control rights from the borrower to the lender at the slightest hint of trouble, especially after having a recent default experience. As a consequence the lender is motivated to write stricter contracts (Murfin (2012)).¹ Stricter contracts may lead to changes in a firm’s investment policy by changing the incentives of the manager due to the changed nature of the allocation of control rights. The manager may choose an investment policy that minimizes the probability of a future covenant violation and the subsequent transfer of control rights to the lender. Additionally, lenders learning from defaults, enables them to assess future prospects better. The lenders may update their beliefs on investment opportunities and thus discourage borrowers from investing in over optimistic projects.

Second is a ‘borrowing cost’ hypothesis, where the lender with recent default experience externalize their losses by charging higher interest rates on its subsequent loans, leading to a higher cost of capital for borrowing firms. In order to overcome the negative profitability shock of the default, lenders prefer to charge more for a given level of credit risk. Also, if they discover any weakness in their screening and monitoring processes, rectifying them will also incur costs, which inflates the interest rates charged to new borrowers. In the syndicated loan market, corporate defaults may harm the reputation of the lead arranger, making them unattractive for other lenders to join their syndicate (Gopalan et al. (2011)). Therefore, the lenders with recent default experience may charge higher loan spreads along with retention of a larger fraction of the loans (Gopalan et al. (2011)) and offering stricter contracts to borrowers (Murfin (2012)), so as to facilitate syndicate participation. All else equal, the higher cost of capital may make some profitable projects unviable, thus decreasing corporate investment.

However, there is one possibility that is not considered in the above hypotheses that can potentially explain why borrowers invest lesser when their lenders face recent default experience. To the extent that firms with poor investment opportunities borrow from lenders facing capital constraints, any negative association

¹In untabulated tests, we examine how strictness of the contract is influenced by recent default experience. Specifically, we use *Covenant Slack* as the dependent variable in a specification similar to Panel A of Table II and find that recent default experience is negatively associated with covenant slack, confirming the findings of Murfin (2012). We define *Covenant Slack* as the difference between initial current ratio (net worth) and covenant specified threshold of current ratio (net worth) scaled by the standard deviation of current ratio (net worth). By construction, a higher value of *Covenant Slack* indicates a lower lender specified threshold of current ratio (net worth), or in other words lender leniency. Dividing by the standard deviation of current ratio (net worth) allows standardized comparison across different *Covenant Slack* measures. In case of multiple *Covenant Slack* measures in a single loan, we use the minimum measure so as to focus on the most strict covenant.

can be simply due to the endogenous matching between lenders and borrowers (Schwert (2018)). We perform a series of tests to rule out this potential explanation in subsequent sections. Specifically, we control for lender characteristics, examine changes in investment behavior, perform panel data regressions, and examine cross-sectional variation in the main findings according to borrower, lender, loan, and lender-borrower relationship characteristics.

Using data between 1987 and 2016 on bilateral and syndicated loans from Dealscan and data on corporate credit ratings from Compustat ratings database, we first identify the lenders with recent default experience. Next, we identify the firms that borrow from such lenders facing defaults. We find that these borrowers are more likely to have a pre-existing relationship with the lender with default experience (i.e., lend to relationship borrowers) and approximately on average pay 5.6% higher spreads (coefficient of 0.056 on *Lender Shock - 90 Days* and mean log spreads (*AIDS*) of 5.2874 in Table II). These findings suggest that when faced with defaults, lenders face stronger incentives to exploit familiar borrowers within their portfolio.

Next, we examine the quarterly investment activity at the new borrowers and find that firms that borrow from a lender with a recent default reduce investment by 7.41% after controlling for the standard determinants of investment. The economic effect when compared to the effect of one standard deviation changes in *Q*, *Cash Flow*, and *Firm Size* of 25.76%, 4.09%, and -15.99%, respectively, is economically large and comparable to the standard determinants of investment. We also find that this negative effect on investment persists for at least two years after the default event.

To examine the robustness of our findings, we perform a variety of tests. First, to rule out the possibility that these findings are due to capital constraints faced by the lenders or because of macroeconomic conditions, we control for various measures of lender capital constraints and macroeconomic variables. Controlling for lender capitalization allows us to examine the role of lenders on firm investment policy unconditional of default experience and also to examine whether lender portfolio defaults have any residual effect on borrower investments beyond capital constraints. To obtain lender characteristics, following Keil (2018), we run a parsing, matching, and filtering algorithm to match lenders in Dealscan to federal databases on banks to obtain comprehensive information on lender characteristics (refer to Section II for more details on the matching) and obtain measures on lender's book assets, capitalization, and risk-adjusted capital ratios. The inclusion of lender controls in our investment regressions shows that lender default experiences still influence borrower investment beyond its effect on lender capitalization. However, when we partial out the independent effect of recent default experience, we find that firms that borrow from well-capitalized lenders invest more. Thus, lender capitalization is another channel through which supply side frictions influence borrower's investment unconditional of lender recent default experience. For measures of macroeconomic conditions, we adopt those used in Murfin (2012) and find that *Lender Shock* has a persistent effect on

investment, even after controlling for macroeconomic conditions. Specifically, the persistence of *Lender Shock* when we control for macroeconomic conditions suggests that the negative effect on borrower investment comes from idiosyncratic default experience and not something that lenders can learn vicariously from other lenders experiencing defaults in their portfolio.

Second, if the decrease in investment is a result of endogenous matching between lenders and borrowers, then there is no reason to expect the changes in borrower’s investment to be related to the default experience although the level of investment can be explained by it. We examine the relationship between changes in investment (with respect to the quarter before the loan) and lender’s recent default experience. Alternatively, we also perform panel data regressions using a sample of borrowers for the duration of their loan contracts. In both specifications, we find that our main results remain robust.

Third, to rule out the possibility that both defaults and investment opportunities can be correlated across regions or industries, we recompute the *Lender Shock* measure by excluding defaults from the same industry or the same U.S. state, and reestimate our regressions. Our results remain unchanged, suggesting that the decrease in investments is not because of lenders concentrating within an industry or geographic region.

Finally, we examine cross-sectional variation according to loan, borrower, relationship, and lender characteristics, and according to changes in creditor markets. For example, if lenders with recent default experience make more risky loans, or lend more to high agency cost borrowers, or lend exclusively to less familiar borrowers, or lend under dire lender circumstances, we expect to find a stronger effect on investment in high risk loans, in loans to risky and unfamiliar borrowers, and when weaker lenders make these loans. However, we find that our results are robust across all subsamples, irrespective of any of the characteristics of the loan, the borrower, the relationship, or the lender. These findings support the ‘learning’ hypothesis. Recently, the participation of non-bank lenders in the corporate lending market has increased as witnessed by the proliferation of covenant-lite loans and increase in institutional participation. Such changes have lowered the necessity and importance of bank monitoring of corporate loans (Becker and Ivashina, 2016). Consistent with such an evolution in the credit markets, we find that our results are weaker for covenant-lite loans and loans with high institutional participation, suggesting that bank incentives to monitor moderate the relationship between recent default experience and corporate investment. We also find that our results are weaker during financial crises periods, further supporting the ‘learning’ hypothesis, as systemic defaults contain less information about screening abilities and processes of the lenders.

We also investigate how the recent default experience of a syndicate participant affects borrower’s investment. Syndicate participants usually have comparable economic exposure to the loan as the lead arranger, and if lenders are highly connected to each other in the form of small and closed syndicates, a default experience of a syndicate member can also have a ‘learning’ effect on any new lending. Alternatively, if lenders

are largely independent, forming different syndicates over time, then it is not necessary that a default experience of one particular syndicate member affects the contractual arrangement in subsequent lending. In our analysis, we find that a shock to a syndicate participant also has a similar effect on borrowing firm's investment.

Chava and Roberts (2008) show that financial covenant violations are negatively associated with firm investments. If firms that borrow from lenders with recent default experience receive stricter contracts on average (Murfin (2012)), then it is possible that actual covenant violations are the channel through which investments decrease. In other words, lender default experience influences the covenant thresholds, and only affects the investment through covenant violations as documented in Chava and Roberts (2008). On the other hand, if covenant violations are not related to lender's recent default experience, it suggests that there are other potential ways through which lenders restrict the borrowing firm's investment. To investigate how default experience affects the probability of covenant violation, we estimate quantitative financial covenant violations including both current ratio and net worth covenants by comparing covenant thresholds in the loan contracts with quarterly financial reports. We find only weak evidence that lender's recent default experience is positively associated with future covenant violations. Also, examining the time to first covenant violation, we again find weak support that borrowers from lenders with recent default experience violate covenants sooner. Thus, these findings suggest that actual covenant violations may be an alternative channel through which firm investment decreases. However, our main finding of the immediate decrease in firm investment following the loan (i.e., in the quarter after the loan contract) suggests that covenant violations cannot explain the entire decrease in investment, but only provides complementary evidence how lender-borrower nexus can impact investment.

Our paper is related to the literature investigating frictions to loan supply and its real consequences. Murfin (2012) documents the tightening of loan contracts when lenders are faced with recent default experience. Similarly, Gopalan et al. (2011) investigate the effect of large-scale bankruptcies on lead arranger's subsequent syndication activity, and find that the lead arranger encounters difficulty in attracting other lenders to form the syndicate. To mitigate the effect of bankruptcy on reputation, the lead arranger offers tighter contracts and retains larger fraction to attract syndicate participants. Our paper is also related to studies that examine how different kinds of shocks affect loan supply. For example, Bord, Ivashina, and Taliaferro (2015) examine how lenders exposed to counties that suffer adverse real estate shocks contract their loan supply to borrowers in different unaffected counties. Also, Chava and Purnanandam (2011) show that banks exposed to the 1998 Russian sovereign default shrink their lending. We extend these studies by documenting a new channel by which lenders can propagate their individual default experience to the economy. Our results also highlight who bears the costs of mistakes of bankers, further propagating the

too big to fail view. Finally, our paper is also related to studies that investigate the link between financing and investment and specifically show how financing market frictions in the debt market affects investment (Whited (1992); Hennessy (2004); Chava and Roberts (2008)). Our paper complements these studies and identifies the effect of a new source of friction in the debt markets on investment.

The remainder of the paper proceeds as follows. Section II describes our data and sample construction. Section III presents our empirical findings examining the impact of recent default experience on investment. Section IV discusses the cross-sectional variation in the investment response to default experience, investment response to syndicate participant’s default experience, and the effect of default experience on probability of covenant violations. In Section V we conclude.

II. Sample construction

A. Identification of recent default experience of lenders

To examine the effect of lender default experience on corporate investment, we begin our empirical analysis by identifying recent default experiences of lenders between 1987 and 2016, following Murfin (2012).² This paper focuses on actual defaults instead of technical defaults like covenant violations as in Chava and Roberts (2008), for the following reasons. First, a technical violation like a covenant default is less likely to affect the profitability of the lender, and thus be less powerful in inducing any changes in their lending behavior to unaffected borrowers. Second, setting up of covenant thresholds is endogenous to the level and quality of monitoring desired by the lender. For example, a lender might choose higher (lower) loan spreads and lower (higher) covenant thresholds when they (do not) specialize in monitoring and interventions. Therefore, if we choose covenant violations as our source of shock, then our sample might be biased towards banks that specialize in active monitoring. Finally, Denis and Wang (2014) show that lenders and borrowers actively renegotiate covenants before their actual violation, which suggests that observed covenant violations might be understated and could systematically bias our sample in favour of lenders facing higher agency costs or lenders who are unwilling to actively monitor the firm i.e., prefer to behave like bondholders.

To identify recent default experience, first we create a list of firms that are reported to be in default (rating ‘D’) or selective default (‘SD’) in the Standard and Poor’ (S&P) Compustat ratings database that is updated at a monthly frequency.³ These firms are reported to miss or delay one or more payments on

²We begin our sample period in 1987 to coincide with the availability of S&P rating data and a sufficient sample of Dealscan loans. However, in later tests on covenant violations and credit market changes, we restrict our attention to sample periods beginning in 1994 and 1997, respectively, to coincide with the availability of reliable covenant data (Chava and Roberts, 2008) and beginning of leveraged loans (Lim, Minton, and Weisbach, 2014).

³S&P rates a borrower ‘SD’ (selective default) or ‘D’ (default) when the borrower has failed to pay one or more of its financial obligations on either rated or unrated securities when the payments came due. A ‘D’ rating indicates that S&P believes the borrower will fail to pay all or substantially all of its obligations when they are due. An ‘SD’ rating indicates that S&P believes

any of their outstanding financial obligations. The S&P Compustat ratings database records all publicly listed firms in the US with a credit rating. Typically, these firms tend to be large and have access to alternative sources of funding, e.g., access to public bond markets. Specifically, during an economic crisis or contractions in the money supply, these firms have better access to alternative sources of financing (Becker and Ivashina (2014)). With this caveat in mind, using this rich data provides us an empirical set up to test more meaningful idiosyncratic shocks in the form of corporate loan defaults of large borrowers to a new borrowing firm’s investment behavior (Murfin (2012)).

Second, we merge these borrowers in default with the Loan Pricing Corporation’s Dealscan (Dealscan) loan database which consists of syndicated and bilateral loans made by financial institutions to publicly held nonfinancial firms. Different measures in the Dealscan data are available in different granularity. A tranche or facility is the most basic unit of observation, which is usually aggregated into a deal or package.⁴ A facility refers to a specific loan amount and maturity available in the borrowing package. For example, a firm that enters into a revolving credit facility can be provided two or more tranches with different maturities and different amounts as part of a single loan package. Each facility in Dealscan consists of information including terms of the deal like the loan size, interest rate spread, covenants, maturity, and other lending details, which may be the same within a loan package.⁵

[Figure 1 about here.]

Figure 1 plots the quarterly distribution of identified default shocks and the distribution of affected lenders during our sample period from 1987 to 2016, when a lender has experienced a default shock in the past 90 and past 360 days prior to the new loan issuance, respectively. Our data consists of a large number of defaults that occur after the burst of the “Dotcom Bubble” and prior to the “Great Recession”, presumably during an era of high credit expansion. Unsurprisingly, the 360 day shocks are distributed more smoothly by virtue of their construction. Cross-sectional differences in lender exposure to default shock over the calendar quarter and calendar year help us to identify the role of these idiosyncratic shocks on corporate investment of new borrowers in Section III. The distribution of these aggregate shocks clearly show that idiosyncratic lender shocks occur even during the periods with relatively better economic performance, mitigating the concern that these shocks might simply capture business cycle fluctuations during our sample period.

that the borrower has selectively defaulted on a specific obligation, but they will continue to meet their obligations on other issues or classes of obligations promptly.

⁴A loan facility or tranche is the basic unit of observation in most studies that examine a borrowing firm’s financial obligations from Dealscan database (e.g., Chava and Roberts (2008); Roberts and Sufi (2009)).

⁵When performing analysis of covenant violations, we aggregate the facility level information to package level by observing the maximum package maturity and the stricter current ratio and net worth covenants in case of multiple covenants (Chava and Roberts (2008)).

B. Identification of new borrowers according to lender default experience

To examine whether the merging of Compustat and Dealscan data introduces any bias, we first compare key firm characteristics before and after the merge. The comparison is presented in Panel A of Table I. In the first three columns, we provide the summary statistics of firms in the Compustat quarterly database, i.e., Compustat sample after excluding financial firms (i.e., SIC codes 6000-6999) for a total of 13,560 unique firms comprising of 507,592 observations. In the next three columns, we provide the summary statistics for the merged, i.e., Compustat-Dealscan sample for a total of 4,014 unique firms comprising of 15,734 observations. Comparing the two samples, we can see that firms in the two databases are typically larger, highly levered, more profitable, have higher net worth and lower current ratios, have fewer growth opportunities, invest lesser, and face higher bankruptcy risks. The bias towards larger and more mature firms in the merged sample is consistent with lender preferences for a typical borrower (Schwert (2018)).

[Table I about here.]

Next, using Dealscan database, we identify the lead lenders with recent default experience as explained in Section II.A. In the empirical analysis that follows, we mainly focus on the new loans that are originated by these lead lenders with default experience in the subsequent two (i.e., most recent quarter) to five quarters (i.e., most recent year) following default and compare them with those lead lenders who provide new loans but have no recent default experience. When identifying the lenders with recent default experience, we do not include the loans originated immediately in the quarter following the default (i.e., not included in the treated sample) as preparing the loan tear sheets and the syndication process on average takes a calendar quarter (Murfin (2012)). Specifically, loans originated by a lead lender who faces default in the previous quarter is included as a new loan from a lead lender without a recent default experience (i.e., included in the control sample). The default shock is unlikely to modify lender behavior in the subsequent immediate quarter when potential new loans are in advanced stages of negotiation already. Also, we limit the timing horizon to five quarters following the default as the effect of the default experience will become a distant memory for the lender after a year from the default, and thus less likely to modify lending behavior after a year. Furthermore, regular changes to lending practices and turnover of loan officers suggest that defaults that happened in the distant past might be less informative or relevant to the bank's current screening ability.

In the last three columns in Panel A of Table I, we provide the summary statistics for the subsample of firms in the Compustat-Dealscan merged sample that has borrowed from a lender who had experienced at least one default in the most recent quarter (ignoring the immediate quarter after default as described

above).⁶ The subsample consists of 468 unique firms comprising of a total of 800 observations. Comparing the subsample to the Compustat and the Compustat-Dealscan merged samples, we observe that most of the borrower firm characteristics are similar across subsamples.

C. *Loan and lender characteristics*

Murfin (2012) shows that lenders impose significantly stricter contracts on borrowers when they experience a recent default. In Panel B of Table I, we compare the loan characteristics in the whole Compustat-Dealscan sample, to the default and non-default experience subsamples. We find the mean loan spread is higher on average when firms borrow from the lenders with recent default experience. The loan maturity tends to be shorter, with larger loan amounts, and more number of participants in the loan syndicate in the subsample of lenders with recent default experience. These observations on the sample data tentatively suggest that the loans from the lenders with recent default experience might be considered riskier than their counterparts. However, these findings do not differentiate between borrowers being more risky or lenders being more fragile.

In Panel C of Table I, we attempt to address this issue in a univariate setting by comparing the characteristics of lenders based on whether or not they have recent default experience.⁷ To the extent that poorly capitalized lenders face more default shocks, then their borrowers' investment behavior can simply be a manifestation of borrowers with poorer investment opportunities matching with poorly capitalized lenders, thus making borrower investments to be an endogenous outcome of the matching process (Schwert (2018)).⁸

To obtain lender characteristics without losing a significant amount of observations, we first link the Dealscan loans data to Federal Deposit Insurance Corporation (FDIC) database, and second to Federal Reserve / National Information Center database. The FDIC and FED identifiers, i.e., RSSD allows us to link the lenders with the facility identifier in the Dealscan database using a parsing, matching, and filtering algorithm proposed by Keil (2018).⁹ We confirm that this matching procedure is comprehensive enough by including the lender link table provided by Schwert (2018). Results in Panel C of Table I suggest that the lender capitalization and Tier-1 capital ratio is lower among lenders with recent default experience, which support the argument of Schwert (2018). However these lenders also have a significantly larger amount of

⁶For all analysis in Table I, we use the most recent quarter to classify lenders into those with recent default experience and those without (i.e., *Lender Shock 90 Days*). But in later tests for robustness, we also use an alternative definition of recent default experience by examining those lenders who face default in the previous year (i.e., *Lender Shock 360 Days*).

⁷As documented in the prior literature, bank capitalization and lending activity is significantly associated with each other (Cornett, McNutt, Strahan, and Tehranian (2011); Beltratti and Stulz (2012); Berger and Bouwman (2013); Carlson, Shan, and Warusawitharana (2013)). Therefore, we use some of the important measures on lender characteristics as control variables in Table IV to control for endogenous lender-borrower matching process as suggested by Schwert (2018).

⁸Murfin (2012) shows that well-capitalized (poorly-capitalized) banks tend to offer less strict (strict) covenants. Schwert (2018) documents that bank dependent (non-bank dependent) firms borrow heavily from well-capitalized (poorly-capitalized) lenders.

⁹For more information on the matching procedure, please refer to Keil (2018).

assets with respect to their counterparts in a sample of 821 (13,427) loans with a lender having (not having) experienced a recent default.

D. Borrower characteristics

Using the information on new loans from the lenders with recent default experience, we examine the investing behavior of the new borrowers. For tracking investment behavior, we again use quarterly data as using the highest frequency accounting data can provide the best assessment of how lenders influence borrowing firms' corporate policies. Also, borrowing firms are required to file periodic reports with their creditors detailing compliance with covenants and other loan requirements, usually at the quarterly frequency (Chava and Roberts (2008)). Such additional information provides an opportunity for the lender to provide feedback to reinforce their preferred corporate policies even in the absence of a covenant violation. Panel D of Table I provides mean and median statistics of firm characteristics according to whether a borrowing firm obtains a new loan from lenders with or without recent default experience in a sample of 27,198 and 1,820 firm-quarter observations, respectively.

We find that the borrower's whose lenders have no recent default experience have higher current ratios, lower net worth, higher growth opportunities, and are smaller, and invest more, with respect to borrowers that obtain a loan from a lender who has experienced a recent default shock. These differences in both mean and median are significant at least at the 5% level. Results on the differences in net worth and firm size suggest that firms who borrow from lenders facing recent default experience are not systematically less creditworthy than their counterparts, thereby lowering the concern that our findings are affected by endogenous matching, as discussed in Schwert (2018).

III. Empirical Findings

A. Effect of lender default experience on new lending

After experiencing a default on their portfolios, lenders tend to write stricter loan contracts (Murfin (2012)).¹⁰ Provisions like tighter covenants can ensure that lenders learn about potential trouble at borrowing firms sooner rather than later, and thereby have greater control when borrower performance begins to deteriorate. Bradley and Roberts (2015) highlight trade-offs between covenants and interest rates and

¹⁰Theoretical and empirical evidence strongly suggest a demand-side story, i.e., on average, riskier firms receive strict contracts in the form of number and strictness of covenants (Berlin and Mester (1992); Billett, King, and Mauer (2007); Rauh and Sufi (2010); among others). On the other hand, Murfin (2012) explores the supply-side effect on the borrower-lender relationship and proposes a new measure of contract strictness to capture the ex-ante probability of a forced renegotiation between lender and borrower.

suggest that strict contracts are often accompanied by lower interest rates. However, if a lender experiences a recent default and offers stricter contracts with low interest rates, then inevitably they will face a double-whammy of a negative profitability shock from the default and low interest rates on new loans. Both of our key hypotheses predict an increase in interest rates on new lending following recent default experience. Specifically, when lenders try to fix gaps in their screening and monitoring mechanisms (the ‘learning hypothesis’) and attempt to reverse the negative profitability shocks or try to overcome reputation loss in syndicates (the ‘borrowing cost hypothesis’), we predict that the new loans from lenders with recent default experience are more expensive.¹¹

Bolton, Freixas, Gambacorta, and Mistrulli (2016) argue that relationship lending reduces defaults despite higher initial information acquisition costs, emphasizing the “learning” by banks through their lending activity. Thus, following recent defaults, banks may have greater incentives to lend to a familiar borrower rather than establishing a new relationship with higher uncertainty. From the borrower’s perspective, relationship borrowers face higher switching costs to find new lenders. For example, Bharath, Dahiya, Saunders, and Srinivasan (2009) argues that on average transactional borrowers pay higher borrowing costs when compared to relationship based repeat borrowers, which incentivizes the borrowers to stick to their lenders even when their lenders face recent defaults. Therefore, the lenders can behave opportunistically by inflicting the costs of their default on their relationship borrower locked in due to information asymmetry problems and potential switching costs. Alternatively, when lenders face a default shock, any additional lending to an existing borrower can increase lending portfolio concentration, especially if the risks are correlated across the loan portfolio, such as through common geographic or industry exposure. Thus, the prediction of recent default experience on the choice between relationship and transactional borrower, is ex-ante not clear, and needs to be explored empirically.

Thus both the ‘learning’ and ‘borrowing cost’ hypotheses suggest that when lenders experience idiosyncratic default shocks, they have stronger incentives to charge higher interest rates on the new loans.

[Table II about here.]

In the merged Compustat-Dealscan sample, we examine the interest rates on new loans and the nature of the borrowers according to lender’s recent default experience. The results are presented in Panel A of Table II based on standard errors clustered by firm. In Panel A, the natural logarithm of the all-in-drawn spread is the dependent variable in all the models and the default shock to the lender observed in the recent past is

¹¹It is also possible that lenders use higher interest rates as a screening mechanism to identify good borrowers. Also, higher interest rates may indicate a riskier portfolio of the lender. However, we believe such an explanation is less likely for the following reasons. First, we concentrate on large borrowers because of the nature of our samples where information asymmetry problems are less severe, thereby lowering the need for screening borrowers. Second, the lender faces stronger incentives to lower the riskiness of their total portfolio when experiencing default.

the key explanatory variable. We adopt key control variables from Ivashina (2009) who examines the effect of asymmetric information among syndicate members on loan spreads. To control for industry heterogeneity, business cycle changes in lending behavior, loan characteristics, and borrower creditworthiness, we control for industry fixed effects, year-quarter fixed effects, and include dummies for loan types and S&P credit ratings, respectively. In Models (I)(V), the main variable of interest is represented by an indicator for the shock and in Models (VI)(X), we utilize a cardinal variable as the count of default shocks in the lead lender's portfolio. In Model (I), we find that a lender who has experienced a default in the past 90 days is likely to charge higher interest rates on the loan, where the coefficient on *Lender Shock - 90 Days* is significant at the 1% level. The coefficient estimate suggests that a default shock to the lender portfolio leads to a 5.6% increase in loan spreads to a new borrower. In Model (II), we further control for a default shock that happens between 90 and 180 days in the past. We observe that the coefficient on *Lender Shock - 90 Days* continues to be significant at conventional levels while *Lender Shock 90-180 Days* is insignificant, suggesting that lenders account for these shocks and immediately reflect the new set of information on the newly issued loans.¹² Results remain qualitatively similar in Models (III) and (IV), when we control for default shocks to lenders in the past between 180 and 270 days and between 270 and 360 days periods, respectively. In Model (V), we use an alternative definition of a default shock to the lender by accounting for any shocks that happen during the past 360 days. We find that the coefficient on *Lender Shock - 360 Days* remains significant at a level of 1%. The models using a cardinal variable, i.e., the number of default shocks in Models (VI)(X) produce qualitatively similar results.

In Panel B of Table II we present the results on whether the new lending is likely to be transactional or relationship-based in nature, conditional on the lender experiencing a recent default shock. The dependent variable is an indicator for whether the borrower has had a relationship with the lead bank in the past. Using the specification as in Panel A in which we control for business cycle changes in lending behavior, loan characteristics, and borrower creditworthiness, and additionally controlling for borrower firm fixed effects for the linear probability models, we find that recent default experience is positively associated with the likelihood of the new borrower being a repeat borrower, i.e., positive coefficient in Model (1) on *Lender Shock - 90 Days*, significant at the 10% level. However, in Models (II)(V), the coefficient becomes insignificant, suggesting a weak effect of recent default experience on choice between familiar and unfamiliar borrowers. In Models (VI)(X), replacing the indicator with a cardinal variable, we find that the coefficient on *Lender Shock - 90 Days* is positive and significant in Models (VI), (VII), and (X), at least at the 10% level.¹³

¹²In subsequent analysis, we focus on this efficiency in lenders immediate reaction following default and for the sake of brevity, only report the results using the measure over the 90-day window. Using these alternate definitions in the subsequent analysis does not significantly alter any of our interpretation of the findings that rely on the 90-day window.

¹³This result also complements the findings of Murfin (2012). The stricter contracts offered by lenders with recent default experience could also arise from increased negotiating power that lenders have with their relationship borrowers.

Overall, these findings show that when lenders experience default in their portfolios; they tend to charge higher interest rates on their new loans and prefer to lend to familiar borrowers.

B. Effect of lender default experience on borrower investment: Cross-sectional regressions

Lenders with recent default experience are incentivized to reduce their overall portfolio’s risk and to charge higher interest rates for a given level of risk to overcome the negative profitability shock, to cover the increased fixed costs in improving screening and monitoring systems, and to fix their reputation among syndicate participants. In this section, we provide evidence on how these lender incentives affect investments of the new borrower.

The new loans can have a negative effect on borrowing firm’s investment for the following reasons. First, the ‘learning’ hypothesis suggests that stricter contract terms can explicitly prevent investment in risky projects which usually benefit shareholders more than creditors, i.e., asset substitution problems. Also, lenders could possess significant negative information about specific types of investment opportunities from their default experience and thus forbade the new borrowers in investing in those opportunities. Second, ‘borrowing cost’ hypothesis suggests higher costs of loans can lower the profitability of positive *NPV* projects otherwise and make the potential projects less attractive to pursue. Thus, both the hypotheses predict a negative spillover effect on new borrower investment from the recent default experience.

[Figure 2 about here.]

Figure 2 plots the average quarterly investments around the borrowing. On the left (right) panel, we focus on eight quarters before and after borrowing a loan from a lender with (without) default experience in the previous 90 days. In the left panel, we see the average investments steeply declining and the sharp decline coincides with the start of the new loan. However, on the right panel when borrowers take a loan from lenders without any default experience, we see the average investments marginally increase or stay similar compared to the quarters before the loan. These graphs suggest that the lender’s recent default experience can hurt corporate investment for borrowers. To further investigate this more rigorously, we employ cross-sectional regressions as detailed below.

[Table III about here.]

Using the lender’s recent default experience from the most recent quarter, we perform cross-sectional regressions of new borrower’s quarterly investment activity. We restrict the Compustat quarterly sample to those firm-quarters where the firms have just borrowed a loan from the banks in the Dealscan database. The results are presented in Panel A of Table III based on standard errors clustered by firm. Using investment

defined as capital expenditures scaled by start of period capital stock, as the dependent variable, we include the standard control variables used in prior literature investigating the determinants of investment, e.g., Macro Q, cash flow, and size of the firm (Bond and Meghir (1994); Erickson and Whited (2006); Chava and Roberts (2008)). We also include industry fixed effects to control for industry heterogeneity.¹⁴ In Model (I) we present the results without controlling for conventional investment factors and we find a negative coefficient significant at the 1% significance level. In terms of economic magnitude, a borrower who obtains a loan from a bank with a recent default experience decreases her investment by 12.93% (coefficient of -0.763 divided by the sample mean investment of 0.0590×100 as investment is scaled by 100 in the regressions). In Model (II) and Model (III), we include the control variables in a simple investment regression model, i.e., *Macro Q* and *Cash flow*, and *Firm Size* and results remain significantly negative for *Lender Shock*.¹⁵ Specifically, after controlling for the standard determinants of investment, we find that a borrower who takes a loan from a lender with a recent default experience invests 11.08% and 7.41% lesser on average (estimated coefficient on the variable, -0.654 and -0.437 divided by the sample mean investment of 0.0590×100) in Model (II) and Model (III), respectively. Comparatively, a one-standard-deviation change in the standard determinants of investment including *Macro Q*, *Cash flow*, and *Firm size* leads to an increase in investment by 25.76%, 4.09%, and -15.99%, respectively. Thus, the economic magnitude of the lender’s recent default experience on corporate investment is comparable to the other standard determinants of investment. In Model (IV), we additionally control for *Altman’s Z-score* to account for the effect of financial stability of a borrower on its investment behavior (Altman (1968)). In Model (V), we consider a potential non-linear relationship between a firm’s investment opportunities and its investment and additionally control for the square of *Macro Q*. In Model (VI), we add lagged cash flow to our specification and consider the persistence in the firm’s operating profitability (Chava and Roberts (2008)). All these additional specifications continue to yield robust results with the coefficient on *Lender Shock* remaining negative and significant at the 1% statistical level.

C. Persistence of the effect of lender default experience on borrower investment

In Panels B and C of Table III, we investigate the persistence in the effect of lender default experience on borrower investment. Using the cross-sectional framework in Panel A, we examine the investment activity of borrowing firms in the subsequent periods, i.e., two to eight quarters following the loan issuance. In Panel B, we compare the mean difference in investment between borrowers from lenders with and without default experience, we find that firms in the former group invest consistently lower than those in the latter.

¹⁴Later in additional tests in Section III.F, we include both firm and time fixed effects when we extend our sample to construct a panel dataset.

¹⁵All the variable definitions are detailed in Appendix A.

Although the difference is gradually decreasing over time, it remains strong and persistent as all of the differences in investment between quarters two to eight is statistically significant at the 1% level. To test the extent of the permanent impact of the shock on corporate investment, in Panel C by using the specification in Model (IV) of Panel A, we use investment in two to eight quarters following the new loan issuance as the dependent variable. The timing of the control variables is fixed relative to the measurement period of investment, except for *Lender Shock*, which is time invariant. We find that the coefficient on *Lender Shock* is significant at the 1% level in all the models, suggesting that there is a persistent effect of a lender default experience on borrower investment activity. An intuitive explanation for the persistent decrease in investments is that certain investment activities may happen gradually over time and when such projects are forfeited, the reduction in investment can happen gradually over time giving rise to persistence in investment cuts (e.g., Veracierta (2002) and Khan and Thomas (2008)). This finding also supports the view of Eberly, Rebelo, and Vincent (2012), who attribute the lagged-investment effect to the adjustment cost function of investments (Christiano, Eichenbaum, and Evans (2005)).¹⁶

D. Lender incentives to influence investment

The negative effect of recent default experience on borrower investment can also be due to two alternative explanations: 1) endogenous matching of borrowers with lenders and 2) correlation of business cycle with investments. First, endogenous matching suggests that there can be a potential mechanical relationship between the shrinkage in borrower investment and lender default experience. When banks, especially weaker banks, are hit with a shock that depletes their capital, it is inevitable that they shrink their assets unless they raise new capital or shift to a lower capital to assets ratio (Kashyap, Stein, and Hanson (2010)). So, it is natural to assume that they will lend less and when they do they prefer to lend in the form of smaller loans. Also, endogenous matching (Schwert (2018)) suggests that borrowers with weaker investment opportunities may borrow from a bank with capital constraints, thereby setting up a mechanical relationship between default shock and investment activity at new borrowers. To test this prediction, we control for the lender's capitalization. Specifically, we control for changes in capitalization measured as the ratio of equity to book value (Schwert (2018)), the changes in the lender's risk-weighted *Tier-1* capital ratio (Murfin (2012)), and the changes in the natural logarithm of the lender's total assets (Schwert (2018)).¹⁷ Additionally, we control for the interest rates charged on these new loans, since higher interest rates can indicate a higher level of borrower's risk assessment, in line with the intuition that risky borrowers may be financially constrained and

¹⁶In untabulated results, we test the robustness of our findings to an alternative definition of cash flow as in Eberly et al. (2012) and their double-log transformation of investment equation. Our original findings remain qualitatively unaltered.

¹⁷These control variables also alleviate some of the concerns about the association between bank liabilities and relationship lending (Berlin and Mester (1999)).

be unable to fully finance their investment projects (Fazzari, Hubbard, and Petersen (1988); Almeida and Campello (2007)).

[Table IV about here.]

The results when we control for lender characteristics are presented in Table IV with standard errors clustered by firm. In Models (I)-(IV), we add each of the variable discussed above one at a time to our specification in Model (VII) of Table III. In all the four models, *Lender Shock* remains significant at the 1% level. In terms of these additional variables, we find that Δ *Lender Book Assets* is positively significant in Model (III) and the *All-in-drawn spread* is negatively significant in Model (IV). These findings suggest that when firms borrow from banks with growing asset portfolios and those that borrow at cheaper interest rates can pursue investment projects more aggressively, having partialled out their recent default experience. In Model (V), we include all the controls together and still find that *Lender Shock* remains significant at the 1% level. In Models (VI), (VII), and (VIII), we replace the dependent variable by lead investment in the subsequent periods, i.e., quarter two, three, and four, respectively. We find the coefficient on *Lender Shock* remains negative and significant, except in Model (VIII).

Second, to the extent that defaults in the economy and investment opportunities are correlated, it is natural to find a negative relationship between defaults and future investment. To rule out such an explanation, we control for business cycle variables which can potentially affect the firm’s investment opportunities. Specifically, following Murfin (2012) we control for quarterly GDP growth, credit spreads between Baa-rated and Aaa-rated corporate bonds, quarterly return on the S&P 500, and finally an indicator for the level of aggregate defaults as reported by (S&P) Compustat ratings. Controlling for aggregate defaults in the economy in the same frequency as we measure idiosyncratic default shocks faced by the lenders allows us to account for aggregate macroeconomic risk, allowing *Lender Shock* to capture risks solely from lender’s idiosyncratic lending experience, orthogonal to other common observable macroeconomic risks (Murfin (2012)).

[Table V about here.]

The results when we control for observable macroeconomic variables are presented in Table V. All the specifications are exactly similar to those in Table IV. In all the models, *Lender Shock* remains significant at the 1% level. Additionally, we find that in Models (I)-(IV) when we add these control variables one at a time, all of these additional control variables are significant at the 1% level. Specifically, we find that lower GDP growth, lower credit spreads, higher S&P 500 return, and lower incidence of corporate defaults, are all associated with greater investment by borrowers. *Lender Shock* remains significant when we put all these control variables in a single specification (Model (V)), as well as when we replace the dependent variable with lead investment measured in quarters two, three, and four, respectively (Models (VI)-(VIII)).

The findings in Table IV and V suggest that the negative effect of the lender’s recent default experience on borrower investment is not because of a mechanical relationship between investment and lender capital constraints or business cycle fluctuations.

E. Effect of lender default experience on changes in borrower investment

We further address concerns of endogenous matching between lenders and borrowers in the following ways. First, instead of using the level of investment which may vary between a borrower with and without good investment opportunities, we use the change in investment levels with respect to investment level before the loan issuance. There is less reason to believe that the changes in investment activity in firms are likely to be related to the difference between firms that influence their lender choices. Second, in further analysis, we rely on panel data regressions to control for firm and time fixed effects and examine the validity of our findings obtained from cross-sectional regressions examining the pre-borrowing and post-borrowing investment activity.

[Table VI about here.]

The changes in investment are measured as the difference between investment in the post-borrowing period (from quarters two to five) and the investment in the latest quarter in the pre-borrowing period. The post-borrowing period investment is averaged among the leading quarters. The findings are reported in Table VI with standard errors clustered by firm. We adopt the same specification as in Model (VII) in Table III. Additionally, we control for lagged investment level (e.g., Alti (2006); Eberly et al. (2012)). The dependent variables in Models (I), (II), (III), and (IV) are changes in investment measured with respect to quarters two, three, four, and five in the post-borrowing period, respectively. We find that the coefficient on *Lender Shock* remains significant at the 5% level in all the columns. For example, in Models (I) and (II), the coefficients of -0.225 and -0.199 on *Lender Shock*, when compared to the mean change in investment of -0.0253 and -0.1799 between 2 quarters and 3 quarters, respectively, are substantially large, and can account for 8.89 and 1.11 times the average change between those quarters. In Models (V)(VIII), we replace the indicator *Lender Shock* with a variable for the count of the number of default shocks the lender experiences. Our results remain qualitatively similar to the earlier cross-sectional analysis.

F. Effect of lender default experience on borrower investment: Panel data regressions

As discussed above, we also perform panel data regressions to rule out the possibility that the differences between firms who borrow from a shocked lender and those who borrow from a non-shocked lender drive

our findings. Specifically, using the cross-sectional data on firms that enter into a new loan contract, we construct panel data for the maturity of the loan, i.e., we include quarterly observations for each firm till the maturity of the original loan. With this constructed sample, we then perform regressions with industry fixed effects (firm fixed effects in Model (VII)) and event-quarter fixed effects. Inclusion of event-quarter fixed effects allows us to control for the time-varying effect of the lender default shock on firm investment activity. The results are reported in Table VII with standard errors clustered by firm.

[Table VII about here.]

In Table VII, we replicate the specifications in Models (I)-(VI) of Panel A of Table III. We find that the coefficient on *Lender Shock* is significantly negative at the 1% level in the first six models. In Model (VII) we include firm fixed effects instead of industry fixed effects to control for any time invariant firm level characteristics of the borrower during the loan maturity period. We find that the coefficient on *Lender Shock* is significant at the 10% level. In Models (VIII) and (IX), we replace *Lender Shock* as a count variable (*Shock Count*) instead of an indicator and in Model (IX) we replace the industry fixed effects with firm fixed effects. We find that the coefficient on *Shock Count* is negative in both the models and significant at the 1% level in Model (VIII) but insignificant in Model (IX).

In Panel B of Table VII, we replace the dependent variable from lead investment to investment changes measured in quarters two, the average of two and three, and the average of two, three, and four quarters after the loan issuance compared to the quarter prior to the loan issuance in Models (I)-(III), respectively. In all specifications, we continue to control for industry fixed effects. In Models (IV)-(VI), we include the event-quarter fixed effects additionally. Finally, in Models (VII)-(IX), we replace *Lender Shock* in specifications in Models (IV)-(VI) with the shock count variable. In Models (I)-(VI), we find that the coefficient on our main explanatory variable is negative and significant at least at the 10% level. However, in Models (VII)-(IX) it is insignificant but negative.

Taken together, the findings in Table VII reinforce our main findings and mitigate the concern that endogeneity of borrowers drive our results.

IV. Robustness Tests

A. Effect of industry/region -specific factors on defaults and investment

Another potential concern regarding our finding is that lender defaults and lending preferences may be correlated across industries or geographical regions (Murfin (2012)). For example, a bank may specialize in lending to a particular industry (or region), and the default intensity in the industry (or region) and

investment rate of non-defaulting firms are likely to be correlated, which can raise concerns of spurious correlations. Although in most of our analysis so far, we try to control for unobserved heterogeneity across sample and time, these adjustments will not fully mitigate this concern. Hence, we examine our findings in Table III by excluding new borrowers that belong to the same industry and geographic region as the defaulting firm. Specifically, we redefine the *Lender Shock* and *Shock Count* variables to exclude the defaults that occur in loans that are issued to borrowers in the same industry and region. We use a borrower’s one-digit SIC industry code to define industry relatedness and identify the location of borrower’s headquarters, i.e., state, respectively.

[Table VIII about here.]

The results are presented in Table VIII with industry fixed effects and standard errors clustered by firm. In Models (I) and (III) (Models (II) and (IV)), the dependent variable is the investment in the first (second) quarter following the loan issuance. The control variables are measured with respect to the timing of the investment measure. In both Models (I) and (II), we find that the coefficient on *Lender Shock* is negative and significant at the 1% level. In Models (III) and (IV), when we use the Shock Count instead of *Lender Shock*, we confirm similar results. These findings suggest that the observed effect of lenders’ default experience on borrower’s investment is not driven by spurious correlations that may be due to the specialization of the lender in certain industries or lender’s presence in certain regions.

B. *Cross-sectional variation in findings*

Next, we test whether our findings are sensitive to loan, borrower, relationship, or lender characteristics. Specifically, we test whether there is a systematic variation in firm’s investment sensitivity to default experience along the dimensions of: 1) the type of loan contracts 2) the severity of agency problems associated with a borrower 3) characteristics of the lender-borrower relationship and 4) characteristics of the lender. Our prediction is that if the lender’s effect on investment relates to borrower risk, then the effect of recent default experience will be more pronounced on risky loans, i.e., loans with stricter contracts, borrowers with higher levels of agency problems, and borrowers who are more likely to be first time borrowers with the syndicate. More risky borrowers will receive stricter contracts and have higher levels of agency problems, and consequently their investment will be most sensitive to borrowing from lenders with recent default experience. Similarly, if the negative effect on investment is due to the lender’s fragility, then our results will be more pronounced among lenders with capital constraints. To test for the differential effect on firm investment, we follow Chava and Roberts (2008) and interact the *Lender Shock* variable with a particular proxy for characteristics along the four dimensions discussed above, as specified below:

$$\begin{aligned} \text{Investment}_{i,t} = & \beta_1 I(\omega)_{i,t-1} \text{Lender Shock}_{i,t-1} + \beta_2 (1 - I(\omega)_{i,t-1}) \text{Lender Shock}_{i,t-1} \\ & + \chi_0 I(\omega)_{i,t-1} X_{i,t-1} + \chi_1 (1 - I(\omega)_{i,t-1}) X_{i,t-1} + \epsilon_{i,t} \end{aligned} \quad (1)$$

where $I(\omega)$ is an indicator function that takes the value of one for specific subsample according to the different proxy measures and zero otherwise. Estimating the above equation is better than estimating our baseline specification over different subsamples as it allows easy comparison of the coefficients in the same model as the errors are not interacted with the indicator function (Chava and Roberts (2008)).¹⁸

For measuring loan contract terms, we choose proxies from the prior literature (Bradley and Roberts (2015); Chava and Roberts (2008); Ivashina (2009)). First, risky loans are going to be costly. To that extent, we create an indicator variable for the *All-in-drawn Spread*, which takes the value of one if the spread is above the sample median and zero otherwise. Second, lenders are also more likely to impose financial covenants that transfer control to the creditors from the management in case of covenant violations. Therefore, we define an indicator variable for *Financial Covenant* that takes the value of one for loans that include one or more financial covenants and zero otherwise. Third, lenders prefer to lend on a *Secured* basis as opposed to unsecured, especially when they consider the loan to be risky. So, we define an indicator *Secured* to indicate the nature of the loan. Fourth, when making risky loans, lenders prefer to lend for shorter maturity. Hence, we define our fourth proxy *Maturity*, which takes the value of one for loans with above median maturity in our sample and zero otherwise.

For measuring agency costs, we choose the following proxies based on prior studies that examine agency problems in the context of a firm's contractual environment (Jensen and Meckling (1976); Ashbaugh-Skaife, Collins, and LaFond (2006); Almeida and Campello (2007); Nikolov and Whited (2014)). Agency problems can be more severe when firms rely less on external capital and when information asymmetry problems between managers and investors are severe. To capture these dimensions of agency problems we use following measures: 1) whether the firm is financially constrained or not according to the *Whited-Wu* index (WW-Index); 2) whether the firm has above median cash holdings (*Cash Holding*); 3) whether the firm has a credit rating or not (*Rated*); and 4) the ratio of fixed assets to total assets which can increase debt capacity of the firm (*Tangibility*).

For measuring whether the borrower has had a prior relationship with the lender, first we observe whether the current lead lender has been a past lead lender for the same borrower in the Dealscan data on syndicated

¹⁸In additional untabulated tests, to examine the robustness of our findings to alternate specifications, we also estimate the regressions using two subsamples divided using the $I(\omega)$ function and then compare the coefficients on *Lender Shock* across the two subsamples. All the results of test of difference in coefficients are qualitatively similar to those presented in Tables reftab:table9 and 10, and are available upon request.

loans (Rosenfeld (2014)).¹⁹ We characterize such a relationship as “*S.Relation Lead Lender*”. Second, to account for the possibility that the current lead lender could have had an indirect relationship with the borrower by participating in a syndicate that provided a loan in the past, we define “*A.Relation Lead Lender*” as those pairs, where the borrower has borrowed a loan in the past from a syndicate which included the current lead lender as a participant. Third, to see if at all there is any overlap between the current syndicate participants and any syndicate participants in a past borrowing, we define “*A.Relation Syn. Member*” as those pairs where at least one syndicate participant is common between the current loan and a past loan. Finally, to account for the home bias of lenders (Carey and Nini (2007)), we create a distance measure between the lead lender and the borrower, “*Distance Lead Lender*”. The borrowers who are nearer to the lead lender are more likely to be familiar to the lender and thus suffer less severe information asymmetry problems from a lender’s perspective.

For measures of lender characteristics, we borrow the measures from Schwert (2018) as used in Table IV. Specifically, we create subsamples based on the level of lender capitalization measured as the ratio of equity to book value, the changes in lender capitalization, the changes in the lender’s risk-weighted *Tier-1* capital ratio, and the changes in the natural logarithm of the lender’s total assets (Schwert (2018)).

[Table IX about here.]

The results using these variables and the specification in Eq. (1) above is presented in Table IX, with industry fixed effects and standard errors clustered by firm. For the sake of brevity, we only present the coefficients on $I(\omega)$ *Lender Shock* and $(1 - I(\omega))$ *Lender Shock*. We also present the F -statistics of the test of difference between these two coefficients and the associated p -value. In Panels A, B, C, and D, we use the proxies for loan contract terms, agency costs, relationship characteristics, and lender characteristics, respectively. In Panel A, we find that both the coefficients are negative in all columns using *All-in-drawn Spread*, *Financial Covenant*, *Secured*, and *Maturity*. The coefficients are also mostly significant, except in two cases including the loans without financial covenants and loans with longer maturity. The F -statistics for the test of difference in the two coefficients is also not significant in any model. These findings suggest that the sensitivity of borrower investment to recent lender default experience does not vary by any loan characteristics (Berlin and Mester (1992); Billett et al. (2007); Rauh and Sufi (2010)). Furthermore, these findings go beyond the contribution of Murfin (2012) in suggesting that even when new borrowers receive lenient contracts, the recent default experience of a lender affects borrower investment. This lends credence to the view that lenders learn about investment opportunities through their recent default experience and hence discourage even the less risky borrowers to invest.

¹⁹Rosenfeld (2014) define a relationship lender as a lender who has been a lead lender in the past five years.

In Panel B as well, we find that all the coefficients on $I(\omega)$ *Lender Shock* and $(1 - I(\omega))$ *Lender Shock* are negative (but insignificant for firms with high tangibility). Also, the F -statistics of the test of difference is insignificant in all the four columns using *WW-Index*, *Cash Holding*, *Rated*, and *Tangibility*. We interpret these results as indicating that the effect of default experience on the borrower is not directly related to the severity of borrower’s agency problems as assumed by standard information models of credit markets, which overweight the demand side characteristics and remain silent about the supply side.

In Panel C, we find again that all the coefficients on $I(\omega)$ *Lender Shock* and $(1 - I(\omega))$ *Lender Shock* are negative (and significant in first three columns), and the F -statistics of the test of difference is insignificant in columns (1) and (4). In columns (2) and (3), although all the coefficients are negative, the test of difference is significant, suggesting that lack of familiarity of the lender with the borrower, has a weaker negative effect of default experience on investment.²⁰ These findings along with those in Panel B, suggest that the negative effect of recent default experience on borrower’s investment is independent of any borrower or relationship characteristics, while lender familiarity with the borrower allows them to identify more investment cuts.

Similarly, in Panel D, using measures for lender capital constraints, we find that the coefficients on $I(\omega)$ *Lender Shock* and $(1 - I(\omega))$ *Lender Shock* are negative for both extremes in lender capital measures. The F -statistics of the test of difference is also insignificant in the last three columns, but significant in the first column. However, the difference in the coefficients in the first column suggest that well-capitalized lenders impose further investment cuts on their borrowers, counter-intuitive to the expected positive relationship between lender capitalization and borrower investment. Thus, these findings suggest that the lender’s capacity constraints do not seem to drive the relationship between the lender’s recent default experience and borrower’s investment.

Overall the findings in Table IX suggest that our baseline findings of a negative effect of lender’s recent default experience on borrower investment activity cannot be explained through observable loan contract terms or through the characteristics of the borrower or the relationship or the lender. The effect seems to be more universal and coming exclusively from the lender’s idiosyncratic experience and an update of their beliefs and lending processes, strongly supporting the ‘learning’ hypothesis.

²⁰In the second and third columns, the significant F -statistics suggests that the borrowers who borrow from a syndicate with one or more common syndicate lead or participants when compared to a past loan, suffer greater reduction in investment activity when compared to borrowers who have no overlap among syndicate participants. Although, the coefficient on both kinds of syndicates are negative and significant, the significant difference between the two coefficients suggests that when any of the syndicate member is familiar with the borrower through a past transaction, they are able to better identify or propose investment cuts.

C. Evolution in the lending markets

The first decade in the twentieth century saw some significant structural developments in the corporate lending market, with non-bank institutional investors participating and playing increasingly larger roles in corporate lending (Lim et al., 2014). The increased participation of such investors broadened the syndication supply and narrowed the skills and muddled the incentives of the lenders (Becker and Ivashina, 2016). These developments also seemed to have significantly altered the nature of contracting in the corporate lending market, with covenant-lite loans becoming more popular, as a result of the increased coordination costs of the institutional investors. Such changes in the market affect the role of a lender and their incentives to actively monitor the borrower. They face greater incentives to cater to the institutional demand to earn a fee income, instead of thoroughly screening and monitoring the borrowers.²¹

To examine the effect of such changes in the credit markets we first select suitable measures that track these developments. First, we classify loans into leveraged and non-leveraged. Although there is no universal definition of a leveraged loan, Dealscan defines a leveraged loan as a syndicated loan that is unrated or rated at BB+ or lower and has a spread over 150 basis points over benchmark rates. Second, we classify a loan as being covenant-lite, based on Dealscan market segment file, which provides a tag for “Covenant lite” (Berlin, Nini, and Edison, 2020). Third, we classify loans as either having or not having institutional investor participation based on the primary SIC code of the lenders. Specifically, we categorize a loan to have institutional participation if at least one non-commercial bank institutional investor is present in the syndicate (Lim et al., 2014).²² Finally, we construct an indicator for financial crises to delineate loans that were issued in the aftermath of the “Dot-com Bubble” and the “Great Recession”.

[Table X about here.]

We use the specification in Eq. (1) to perform the cross-sectional variation analyses of our findings according to these measures of changing credit market characteristics. The results are presented in Table X, with industry fixed effects and standard errors clustered by firm. For the sake of brevity, we only present the coefficients on $I(\omega)$ *Lender Shock* and $(1 - I(\omega))$ *Lender Shock*. We also present the F -statistics of the test of difference between these two coefficients and the associated p -value. In the first column using an

²¹Responding to a letter by Senator Elizabeth Warren in May 2019 that raised concerns of overheating in the leveraged loan markets, the Office of the Comptroller of the Currency (OCC) Joseph Otting, Fed Chair Jerome Powell, and FDIC Chair Jelena McWilliams replied “Agency examiners have observed in some transactions fewer and less stringent protective covenants, more liberal repayment terms, and incremental debt provisions that allow for increased debt that may inhibit deleveraging capacity and dilute repayment to senior secured creditors, acknowledging the potential risks. Furthermore, according to the OCC spokesperson While bank risk management practices are satisfactory, we continue to see risk accumulate, primarily outside of the regulated banking system where there is much less transparency, making it more difficult to monitor. The OCC has been discussing with bank management and directors the need to consider the potential effect on the financial system from originating and distributing weakly underwritten loans to leveraged borrowers.

²²A lender is classified as a commercial bank if its primary SIC code falls in 6011-6082 or 6712, and as an institutional investor, otherwise.

indicator for leveraged loans, we find that both the coefficients are negative, and insignificantly different from each other. It is possible that some of the leveraged loans classified according to Dealscan definition are traditional loans from lenders that involve close monitoring. Hence, it is not surprising that both kinds of loans reduce their borrower investment. In the next two classification criteria, we try to pin down the changes in the credit markets in a more pointed manner. Categorizing according to covenant-lite status, we find that the negative effect of recent default experience on new borrower investment is only present among non-covenant-lite loans. In column (3), we find that the negative effect is only pronounced among loans without institutional investor participation. Finally in column (4), we find that the negative effect on borrower investment is only present during the non-crises years. The test of difference in coefficients between $I(\omega)$ *Lender Shock* and $(1 - I(\omega))$ *Lender Shock* are significant at least at the 10% level in columns (3) and (4).

Overall, the results in Table X suggest that when lenders are incentivized to learn from their recent default experience such as when they bear the losses or the individual defaults are non-systematic, then they show increased willingness in reducing their next borrower’s investment significantly. These results further support the ‘learning’ hypothesis.

D. Influence of syndicate role

Champagne and Kryzanowski (2007) show that there is a high degree of continuity among syndicates. Lenders face a high degree of information asymmetry with the lead lender in a syndicate, and therefore prefer familiar lead lenders (Champagne and Kryzanowski (2007)) or demand higher interest rate spreads and a greater proportion of loan retention by less familiar lead lenders (Ivashina and Scharfstein (2010)). Under such circumstances, how will the recent default experience of a syndicate participant affect a loan and the borrower’s investment? To the extent that these syndicate networks are constant over time, any default shock to one of the syndicate participants can have the same learning effect as a shock to the lead lender. Furthermore, since the economic impact of default is similar when the lender is the lead or a syndicate participant, the incentives to ensure loan performance remain the same across the syndicate. On the other hand, if these syndicate networks evolve constantly and information asymmetry problems are more severe between less familiar lenders, then a default shock to a particular lender is unlikely to spill over to borrower investment under a different lead lender. We examine these opposing predictions on a recent default experience of a syndicate participant.

[Table XI about here.]

The results are presented in Table XI with industry fixed effects and standard errors clustered by firm.

The main explanatory variables include *Lender Shock* and *Participant Shock*, where *Participant Shock* is an indicator whether a syndicate participant has had a recent default experience. We check the correlation between *Lender Shock* and *Participant Shock* in our sample and find it to be 0.072 significant at the 1% level, which suggests that there is a non-trivial possibility that the lead lenders and syndicate participants have recent default experience at the same time. We replicate all the seven models from Panel A of Table III with an additional explanatory variable for *Participant Shock*. In all the models, we find that the coefficient on *Participant Shock* is negative and significant at the 5% level or better, except Model (IV) where it is insignificant. However, the magnitude of *Participant Shock* is slightly lower than the coefficient on *Lender Shock* in all the models. These findings suggest that even when a syndicate participant member has a recent default experience, the spillover to borrower investment happens in a similar manner as when the lead lender has a recent default experience.

E. Effect of lender default experience on covenant violations

Chava and Roberts (2008) offer evidence that following covenant violations, borrowing firms decrease their investment levels as a result of the increase in the bargaining power of the creditor and the creditor's active intervention.²³ The average time to first covenant violation in Chava and Roberts (2008) sample is 23 months or half the maturity of the average loan. However, in our analysis we find that when a borrower takes a loan from a shocked lender, the negative effect on firm investment is strong in the first few quarters of the loan itself, suggesting that there is a pre-covenant violation effect of the lender on borrowing firm's investment depending on the circumstances facing the lender.²⁴

In order to understand how recent default experience affects the probability of covenant violation and if that can be a channel through which the former affects investment, following Chava and Roberts (2008) we identify all the current ratio and net worth covenants on the new loans.²⁵ Having identified both the

²³Alternatively, Denis and Wang (2014) show that lenders and borrowers engage in active renegotiation prior to covenant violation, often relaxing the limits, in which case firm investment increases significantly, consistent with the view that lenders learn more about their borrowers during the relationship which partially mitigates information asymmetry leading to a better outcome overall. However, in our setting it is unclear about the motivation and probability of engaging in pre-covenant violation renegotiations due to the trade-offs between increased riskiness of the borrower portfolio and the stricter contracts offered to newly issued loans.

²⁴In the context of covenant violations, our findings have interesting and competing implications. If lenders with recent default experience are more likely to monitor the borrowing firms closely and actively intervene in corporate decision making, they can make it happen by offering stricter contracts, thereby increasing the probability of covenant violation and thus obtain greater bargaining power with the borrower. Once the covenant is violated, the creditor can obtain more say in the day-to-day operations, and decrease firm investment further. This argument suggests that borrowing from a lender with recent default experience can have permanent consequences on the borrower, and therefore be very significant to their future. Alternatively, when lenders offer stricter contracts to their new borrowers, they may ex-ante decide to intervene less among their borrowers on covenant violation. Furthermore, managerial attention is also limited when more than one borrower violates their covenants, reducing the probability of close monitoring ex post violation. Such a decision predicts a lower effect of covenant violation on firm investment. Both these views are consistent with our findings of a negative effect on investment by a lender with recent default experience. However, they have opposite predictions for the effect of covenant violation on post covenant violation investment. However, since this is not our key research agenda, we leave this question unexamined and leave it open for future research.

²⁵Covenants are available at the package level but typically applies to each facility as well. Thus, we assume the covenant

current ratio and net worth covenants, we examine whether a firm is in violation at the end of each quarter by comparing the accounting variable for current ratio and net worth in the firm’s financial statements against the threshold specified by the covenant. If the current ratio or net worth falls below the covenant specification, then the firm is estimated to have violated and not otherwise.²⁶

[Table XII about here.]

In Panel A of Table XII, we use an indicator variable for covenant violation as the dependent variable. Adopting the key control variables from Chava and Roberts (2008) and retaining our main explanatory variable *Lender Shock* (or *Shock Count*), we estimate linear probability models with panel data of the probability of covenant violation with standard errors clustered by firm. In Models (I) and (II), we use each type of covenant violation indicator, i.e., current ratio and net worth, estimated quarterly. In Model (III), we use an indicator for any type of covenant violations, using data on annual covenant violations from Nini, Smith, and Sufi (2012). We find that *Lender Shock* is positive in all models and significant at the 5% level for current ratio covenants. In Models (IV)(VI), we replace industry fixed effects with firm fixed effects. However, the coefficient on *Lender Shock* becomes insignificant. The findings in Panel A of Table XI suggest that there may be a weak effect on future covenant violation depending on whether the lender has had a recent default experience.

In Panel B of Table XII, we replace the indicator for covenant violation with the time to covenant violation. We measure the time to covenant violation as the number of quarters in between the loan issue date and the quarter in which the covenant violation is identified as in Chava and Roberts (2008). In the first three models, the specifications remain similar to those in Panel A of Table XII. In the last three models, we replace *Lender Shock* with a count measure. We find all the coefficients on *Lender Shock* is positive, but only significant in Models (I) and (IV), i.e., current ratio covenant violation. This finding suggests that firms that borrow from lenders with one or more default shocks are likely to violate their current ratio covenant sooner rather than later.

When combined with the evidence of Chava and Roberts (2008), our results in Table XII suggest that when firms borrow from lenders with recent default experience, they witness a complete cycle of investment dampening for the duration of the loan contract, brought on by the lender through their lending negotiation, strict contract terms, and increased violation of contracts.

to start at the earliest facility start date and end with maturity, i.e., assume conservatively that firms are subject to strict conditions for the duration of the loan.

²⁶However, there are certain challenges in exactly identifying covenant violations. First, the definition of current ratio and net worth by the creditor can vary according to various adjustments they prefer to be made for the measures. Second, some of the covenants are dynamic, allowing adjustments to be made to the thresholds with time or when the firm makes profits or losses. Third, the borrower might amend the contract thereby making the threshold stale or irrelevant. Chava and Roberts (2008) argue that despite these misgivings, their simple static measure of covenant violation can estimate actual covenant violations quite closely and these challenges do not significantly affect their findings.

F. Robustness to past year shocks

There are some quarters during our sample period when no bank in our sample experiences a recent default shock in the previous quarter. This raises the concern whether our findings are driven by some omitted time variant characteristics. To further mitigate this concern, we re-estimate our regressions using a recent default shock experienced in the previous year instead of the previous quarter. As can be seen in Figure 1, the annual default shocks are much more smoothly distributed across our sample period.

[Table XIII about here.]

The results are presented in Table XIII with industry fixed effects and standard errors clustered by firm. In Models (I)-(IV) (Models (V)-(VII), the dependent variable is the investment in the first (second) quarter following the loan issuance. The control variables are measured with respect to the timing of the investment measure. In both Models (I) and (II), we find that the coefficient on *Lender Shock* and *Shock Count* is negative and significant at the 1% level, respectively. In Models (III) and (IV), we add all the lender control and macroeconomic control variables used in Table IV and Table V, respectively. The coefficient on *Lender Shock*, continues to remain significant at the 5% level. The results in Models (V)-(VIII) are qualitatively similar when we examine the second quarter investment following loan issuance. Overall the findings in Table XIII relax the time scale of our findings, allowing them to be more generally applicable to the default experience of lenders.

V. Conclusion

This paper identifies a unique channel, namely a lender’s idiosyncratic default experience, through which financing frictions impact corporate investment. Examining the investment activity of a comprehensive sample of US firms, we document that firms that borrow from a lender with a recent default reduce investment by 7.41% compared with their counterparts per quarter after controlling for the standard determinants of investment. These findings are robust to controlling for lender characteristics, macroeconomic characteristics, and when excluding defaulter’s industry and region. Furthermore, the decrease is evident irrespective of loan, borrower, lender, or lender-borrower relationship characteristics. Finally, these findings are more pronounced when lender incentives to learn are in place. We attribute these findings to the idiosyncratic learning experience through which recent defaults lead lenders to update their beliefs about investment opportunities and their own screening ability. Also, the decrease in investments is also consistent with the crowding out effect of higher borrowing costs due to the alteration in incentives of the lender with recent default experience

to expend more resources on screening, to charge higher spreads to attract syndicate participants, and to make up for lost profitability.

Our findings highlight the importance of supply-side determinants of corporate investment activity and consequently the influence on the aggregate economy, differing from traditional theories that emphasize demand-side determinants. Our evidence also improves our understanding of how the contractual environment is affected by lender's idiosyncratic experience and beliefs about economic prospects. Finally, our findings also complement the flight-to-quality and financial accelerator theories (e.g., Bernanke, Gertler, and Gilchrist (1996)) which predict that lenders propensity to discriminate borrowers based on agency costs varies with the business cycle. However, our findings show that when faced with defaults, lenders become conservative irrespective of borrower agency costs or business cycles.

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Table I: Sample Selection and Summary Statistics

This table presents summary statistics (mean, median, and standard error) of samples from Compustat and Compustat-Dealscan merged datasets between the sample period from 1987 to 2016. Panel A presents important variables for three samples of firm-quarter observations. Compustat sample consists of all firm-quarter observations for nonfinancial firms in the Compustat Quarterly files. The Compustat-Dealscan merged sample consists of all firm-quarter observations for nonfinancial firms with one or more new private loans covered by Dealscan. The lender shock sample consists of all firm-quarter observations for nonfinancial firms that have one or more new private loan from a lender who faces a default shock in the past 90 days when the loan is initiated and covered by Dealscan. Panel B presents loan characteristics for three samples of firm-quarter observations in Dealscan. We split the whole Compustat-Dealscan merged sample according to whether the lender faces a default shock in the past 90 days (Lender shock sample) or not (No lender shock sample), when the loan is initiated. Panel C presents lender characteristics for the Compustat-Dealscan merged sample, and subsamples according to whether the lenders face a default shock or not. Panel D presents the mean and the median differences in important borrower characteristics between lender shock and no lender shock samples in the Compustat-Dealscan merged data. Variable definitions are in Appendix A.

Variable	Panel A: Sample Statistics								
	Compustat Sample			Compustat-Dealscan Merged Sample			Lender Shock Sample		
	Mean	Median	S.E.	Mean	Median	S.E.	Mean	Median	S.E.
<i>Current Ratio</i>	2.5054	2.0312	0.0023	1.9211	1.6762	0.0087	1.7783	1.5945	0.0341
<i>Net Worth</i>	254.2730	67.9735	0.5174	379.2853	188.0961	3.2867	469.7770	320.3720	14.4420
<i>Tangible Net Worth</i>	121.4194	41.4570	0.4431	80.4423	59.2540	3.5186	101.8638	96.7575	16.2594
<i>Log(Assets)</i>	5.1160	4.9932	0.0031	6.3179	6.3644	0.0145	6.8506	6.8504	0.0520
<i>Firm Size</i>	3.3933	3.3097	0.0038	4.8308	4.8600	0.0168	5.4054	5.4379	0.0630
<i>MtB</i>	1.8150	1.1928	0.0026	1.3907	1.1214	0.0081	1.2856	1.0759	0.0332
<i>Macro Q</i>	13.2225	5.6524	0.0232	8.3032	3.6668	0.0880	7.2619	3.5527	0.3459
<i>ROA</i>	0.0110	0.0265	0.0001	0.0312	0.0324	0.0003	0.0338	0.0327	0.0009
<i>Tangibility</i>	0.2822	0.2062	0.0003	0.3209	0.2581	0.0019	0.3281	0.2664	0.0086
<i>Investment</i>	0.0643	0.0478	0.0001	0.0590	0.0455	0.0004	0.0511	0.0391	0.0015
<i>Cash Flow</i>	-0.2742	0.0639	0.0022	0.0628	0.0765	0.0035	0.0711	0.0715	0.0132
<i>Leverage</i>	0.2350	0.1979	0.0003	0.3339	0.3108	0.0017	0.3359	0.3183	0.0072
<i>Altman's Z-score</i>	0.6941	0.7139	0.0008	0.6762	0.6378	0.0039	0.6480	0.5656	0.0162

Panel B: Loan Characteristics									
	Compustat-Dealscan			Shock			No Shock		
	Mean	Median	S.E.	Mean	Median	S.E.	Mean	Median	S.E.
<i>All-in-drawn Spread</i>	5.2874	5.4161	0.0053	5.2747	5.4161	0.0224	5.2881	5.4161	0.0055
<i>Loan Maturity</i>	3.7416	4.0943	0.0052	3.7480	4.0604	0.0216	3.7413	4.0943	0.0054
<i>Loan Amount</i>	18.2068	18.4207	0.0140	18.6911	18.8261	0.0505	18.1808	18.4207	0.0145
<i>Log (number of Participants)</i>	1.3586	1.3863	0.0083	1.7189	1.7918	0.0333	1.3393	1.3863	0.0086
<i>Secured</i>	0.7959	1.0000	0.0032	0.7662	1.0000	0.0150	0.7974	1.0000	0.0033
<i>Fixed Charge Coverage Ratio</i>	9.2690	2.5112	0.2344	10.0233	2.5358	1.0848	9.2286	2.5073	0.2401
<i>Lender Shock - 360 Days</i>	0.1780	0.0000	0.0030	1.0000	1.0000	0.0000	0.1339	0.0000	0.0028
<i>Lender Shock - 90 Days</i>	0.0508	0.0000	0.0018	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000

Panel C: Lender Characteristics									
	Whole Sample			Shock			No Shock		
	Mean	Median	S.E.	Mean	Median	S.E.	Mean	Median	S.E.
<i>Lender Capitalization</i>	1.6406	1.6247	0.0055	1.5515	1.6765	0.0169	1.6461	1.6142	0.0057
<i>Lender Tier 1 Capital Ratio</i>	9.3429	8.5500	0.0161	8.9294	8.2000	0.0588	9.3682	8.5700	0.0166
<i>Lender Book Assets</i>	12.9411	13.1214	0.0118	13.7834	13.6122	0.0195	12.8896	12.9831	0.0123

Panel D: Firm Characteristics - Difference									
	Mean				Median - Whole Sample				
	No Shock	Shock	Diff.	S.E.	Wilcoxon	Porder	No Shock	Shock	
<i>Current Ratio</i>	1.9288	1.7783	0.1505***	(0.0396)	0.0000	0.5543	35,530	2,090	
<i>Net Worth</i>	374.4378	469.7770	-95.3392***	(14.9426)	0.0000	0.4106	37,097	2,177	
<i>Tangible Net Worth</i>	79.2948	101.8638	-22.5690	(16.0163)	0.0000	0.4685	37,097	2,177	
<i>Log(Assets)</i>	6.2894	6.8506	-0.5612***	(0.0659)	0.0000	0.4012	37,267	2,184	
<i>Firm Size</i>	4.8001	5.4054	-0.6053***	(0.0763)	0.0000	0.4134	37,090	2,179	
<i>MtB</i>	1.3963	1.2856	0.1107**	(0.0367)	0.0000	0.5300	35,392	2,039	
<i>Macro Q</i>	8.3590	7.2619	1.0971**	(0.4006)	0.0059	0.5189	32,271	1,874	
<i>ROA</i>	0.0310	0.0338	-0.0028*	(0.0013)	0.9810	0.5002	34,394	2,016	
<i>Tangibility</i>	0.3205	0.3281	-0.0076	(0.0087)	0.3072	0.4935	37,161	2,179	
<i>Investment</i>	0.0594	0.0511	0.0083***	(0.0017)	0.0000	0.5424	36,188	2,137	
<i>Cash Flow</i>	0.0624	0.0711	-0.0087	(0.0160)	0.0726	0.5116	35,964	2,124	
<i>Leverage</i>	0.3338	0.3359	-0.0021	(0.0078)	0.0041	0.4817	37,181	2,179	
<i>Altman's Z-score</i>	0.6777	0.6480	0.0298	(0.0176)	0.0000	0.5354	35,413	2,086	

Table II: Characteristics of New Lending According to Recent Default Experience

This tables presents results of OLS regressions of all-in-drawn spreads and linear probability model regressions of likelihood of repeated lending, using loans from Dealscan. The sample consists of loan observations of nonfinancial firms between 1987 and 2016 covered in both Compustat and Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock in the past 90/180/270/360 days when the loan is initiated. Models I-V (Models VI-X) present the results for an indicator for default shock (natural logarithm of count of default shocks) to measure lender shocks. The dependent variable in Panel A is the all-in-drawn spread of the loan contract. The dependent variable in Panel B is an indicator variable that takes the value of one for loans, where the lead lender has had a relationship with the borrower in the past. Industry fixed effects based on 3-digit Standard Industrial Classification (SIC) codes (borrowing firm fixed effects) and year-quarter fixed effects are included in all models in Panel A (Panel B). Other indicators for Loan-type and S&P Long-term Bond Rating are also included. *t*-statistics robust to within-firm correlation and heteroskedasticity are reported in parentheses. Variable definitions are in Appendix A.

Panel A: Loan Spreads According to Recent Default Experience										
	Shock Indicator					Shock Count				
	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII	Model IX	Model X
<i>Lender Shock - 90 Days</i>	0.056*** (2.840)	0.051** (2.478)	0.049** (2.364)	0.048** (2.329)		0.036*** (3.413)	0.033*** (2.924)	0.030*** (2.633)	0.031*** (2.732)	
<i>Lender Shock - 90-180 Days</i>		0.021 (0.986)	0.015 (0.665)	0.013 (0.601)			0.015 (1.273)	0.013 (1.123)	0.013 (1.060)	
<i>Lender Shock - 180-270 Days</i>			0.018 (0.882)	0.015 (0.669)				0.015 (1.331)	0.015 (1.345)	
<i>Lender Shock - 270-360 Days</i>				0.008 (0.377)					0.000 (0.031)	
<i>Lender Shock - 360 Days</i>					0.061*** (4.794)					0.014*** (3.388)
<i>Maturity</i>	-0.024*** (-2.593)	-0.025*** (-2.606)	-0.025*** (-2.615)	-0.025*** (-2.619)	-0.027*** (-2.996)	-0.024** (-2.567)	-0.024** (-2.527)	-0.023** (-2.387)	-0.021** (-2.219)	-0.022** (-2.246)
<i>Amount</i>	-0.073*** (-11.501)	-0.073*** (-11.515)	-0.073*** (-11.517)	-0.073*** (-11.518)	-0.074*** (-11.880)	-0.073*** (-11.502)	-0.073*** (-11.443)	-0.073*** (-11.467)	-0.074*** (-11.568)	-0.074*** (-11.548)
<i>Secured</i>	0.526*** (28.914)	0.526*** (28.901)	0.525*** (28.895)	0.525*** (28.895)	0.529*** (28.719)	0.526*** (28.884)	0.524*** (28.815)	0.524*** (28.719)	0.523*** (28.632)	0.524*** (28.644)
<i>Participants</i>	-0.055*** (-6.692)	-0.055*** (-6.698)	-0.055*** (-6.703)	-0.055*** (-6.701)	-0.055*** (-6.822)	-0.055*** (-6.717)	-0.055*** (-6.688)	-0.055*** (-6.687)	-0.056*** (-6.764)	-0.056*** (-6.749)
<i>Altman's Z-score</i>	-0.104*** (-6.508)	-0.105*** (-6.512)	-0.104*** (-6.511)	-0.104*** (-6.510)	-0.105*** (-6.756)	-0.104*** (-6.511)	-0.104*** (-6.484)	-0.103*** (-6.322)	-0.102*** (-6.362)	-0.102*** (-6.346)
<i>Leverage to Net Worth</i>	0.002** (2.068)	0.002** (2.069)	0.002** (2.051)	0.002** (2.054)	0.002* (1.825)	0.002** (2.065)	0.002** (2.027)	0.002** (2.054)	0.002** (2.010)	0.002** (2.017)
<i>Fixed Charged</i>	-0.001*** (-4.683)	-0.001*** (-4.686)	-0.001*** (-4.694)	-0.001*** (-4.698)	-0.001*** (-4.814)	-0.001*** (-4.659)	-0.001*** (-4.853)	-0.001*** (-4.935)	-0.001*** (-4.863)	-0.001*** (-4.871)
<i>Current Ratio</i>	-0.010* (-1.698)	-0.010* (-1.685)	-0.010* (-1.673)	-0.010* (-1.670)	-0.011* (-1.871)	-0.010* (-1.691)	-0.010 (-1.640)	-0.010* (-1.715)	-0.011* (-1.875)	-0.011* (-1.862)
<i>Tangible Net Worth</i>	-0.063*** (-8.824)	-0.063*** (-8.830)	-0.063*** (-8.820)	-0.063*** (-8.814)	-0.064*** (-9.120)	-0.063*** (-8.834)	-0.064*** (-8.902)	-0.063*** (-8.844)	-0.064*** (-8.854)	-0.064*** (-8.844)
<i>Intercept & Other Indicators</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry & Year-Quarter FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Obs.</i>	14459	14459	14459	14459	14983	14459	14333	14190	14028	14028
<i>Adj.R-square</i>	0.563	0.563	0.563	0.563	0.563	0.563	0.564	0.566	0.570	0.570

Table II: Continued

Panel B: Likelihood of Repeat Borrower According to Recent Default Experience										
	Shock Indicator					Shock Count				
	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII	Model IX	Model X
<i>Lender Shock - 90 Days</i>	0.042*	0.026	0.010	-0.005		0.034**	0.028*	0.020	0.014	
	(1.921)	(0.821)	(0.303)	(-0.144)		(2.107)	(1.660)	(1.159)	(0.822)	
<i>Lender Shock - 90-180 Days</i>		0.027	0.019	0.013			0.025	0.016	0.010	
		(1.639)	(1.128)	(0.781)			(1.509)	(0.992)	(0.615)	
<i>Lender Shock - 180-270 Days</i>			0.046***	0.039**				0.044***	0.037**	
			(3.053)	(2.568)				(2.930)	(2.441)	
<i>Lender Shock - 270-360 Days</i>				0.043***					0.042***	
				(2.833)					(2.783)	
<i>Lender Shock - 360 Days</i>					-0.008					0.027***
					(-0.485)					(4.401)
<i>Maturity</i>	-0.024**	-0.022***	-0.022***	-0.023***	-0.028***	-0.024***	-0.022***	-0.022***	-0.023***	-0.023***
	(-2.624)	(-2.718)	(-2.685)	(-2.829)	(-3.588)	(-2.958)	(-2.697)	(-2.666)	(-2.807)	(-2.757)
<i>Amount</i>	0.026***	0.026***	0.025***	0.024***	0.027***	0.026***	0.026***	0.025***	0.024***	0.024***
	(4.917)	(4.673)	(4.586)	(4.338)	(5.154)	(4.760)	(4.640)	(4.550)	(4.295)	(4.271)
<i>Secured</i>	-0.002	-0.003	-0.001	0.000	0.003	-0.002	-0.003	-0.001	0.000	0.000
	(-0.080)	(-0.140)	(-0.040)	(0.004)	(0.173)	(-0.090)	(-0.138)	(-0.038)	(0.009)	(0.003)
<i>Participants</i>	0.057***	0.056***	0.055***	0.055***	0.062***	0.057***	0.056***	0.055***	0.055***	0.055***
	(6.437)	(6.576)	(6.435)	(6.372)	(7.451)	(6.671)	(6.579)	(6.443)	(6.382)	(6.398)
<i>Altman's Z-score</i>	0.007	0.007	0.010	0.011	0.008	0.006	0.007	0.010	0.011	0.010
	(0.306)	(0.296)	(0.407)	(0.432)	(0.326)	(0.255)	(0.288)	(0.396)	(0.418)	(0.374)
<i>Leverage to Net Worth</i>	0.000	0.000	0.001	0.000	0.000	0.000	0.000	0.001	0.000	0.000
	(0.238)	(0.217)	(0.345)	(0.292)	(0.103)	(0.215)	(0.204)	(0.335)	(0.283)	(0.283)
<i>Fixed Charged</i>	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(1.024)	(1.088)	(1.092)	(0.991)	(1.205)	(1.113)	(1.091)	(1.096)	(0.997)	(0.983)
<i>Current Ratio</i>	0.009	0.009	0.009	0.011	0.007	0.009	0.010	0.009	0.011	0.011
	(0.784)	(0.947)	(0.931)	(1.089)	(0.820)	(0.932)	(0.975)	(0.955)	(1.111)	(1.120)
<i>Tangible Net Worth</i>	-0.003	-0.002	-0.003	-0.004	-0.004	-0.003	-0.003	-0.003	-0.004	-0.005
	(-0.234)	(-0.186)	(-0.209)	(-0.305)	(-0.376)	(-0.262)	(-0.210)	(-0.230)	(-0.326)	(-0.364)
<i>Intercept & Other Indicators</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry & Year-Quarter FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Obs.</i>	13914	13777	13622	13454	14541	13914	13777	13622	13454	13454
<i>Adj.R-square</i>	0.226	0.226	0.227	0.227	0.231	0.227	0.227	0.227	0.227	0.227

Table III: Investment Regressions

This table presents results of cross-sectional investment regressions. The sample consists of firm-quarter observations of nonfinancial firms between 1987 and 2016 in the merged Compustat-Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock in the past 90 days when the loan is initiated. All the flow (stock) variables are measured during the fiscal quarter (end of the previous fiscal quarter). In Panel A, the dependent variable is investment in the quarter following the loan. Panel B presents univariate mean differences in investment between borrowers borrowing from shocked and not shocked lenders in two to eight quarters following the loan. Panel C presents the results of cross-sectional regressions of investments in two to eight quarters following the loan. In Panel C, the control variables are measured with respect to whether they are a flow or stock measure with respect to investment, except the shock variable that is measured 90 days prior to the loan. Investment measures are multiplied by 100 for readability. Industry fixed effects based on 3-digit Standard Industrial Classification (SIC) codes are included in all models. *t*-statistics robust to within-firm correlation and heteroskedasticity are reported in parentheses. Variable definitions are in Appendix A.

Panel A: Investment							
	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII
<i>Lender Shock</i>	-0.763*** (0.110)	-0.654*** (0.115)	-0.437*** (0.113)	-0.424*** (0.112)	-0.412*** (0.111)	-0.406*** (0.112)	-0.392*** (0.111)
<i>Macro Q</i>		0.147*** (0.004)	0.129*** (0.004)	0.125*** (0.004)	0.271*** (0.012)	0.123*** (0.004)	0.266*** (0.013)
<i>Cash Flow</i>		0.237*** (0.070)	0.427*** (0.071)	0.210*** (0.073)	0.133* (0.072)	0.151* (0.078)	0.106 (0.077)
<i>Firm Size</i>			-0.395*** (0.023)	-0.317*** (0.024)	-0.333*** (0.024)	-0.310*** (0.024)	-0.325*** (0.024)
<i>Altman's Z-score</i>				1.545*** (0.107)	1.472*** (0.107)	1.493*** (0.113)	1.438*** (0.113)
<i>Sq-Macro Q</i>					-0.003*** (0.000)		-0.003*** (0.000)
<i>Lag Cash Flow</i>						0.176* (0.090)	0.114 (0.090)
<i>Intercept</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Obs.</i>	29398	25221	25221	24213	24213	23796	23796
<i>Adj.R-square</i>	0.155	0.215	0.236	0.257	0.265	0.252	0.260
Panel B: Future Investment - Mean Difference							
	Loan+2	Loan+3	Loan+4	Loan+5	Loan+6	Loan+7	Loan+8
<i>Shock</i>	5.258	5.200	5.168	5.137	5.148	5.078	5.049
<i>No Shock</i>	6.016	5.939	5.764	5.799	5.667	5.635	5.505
<i>Difference</i>	-0.758	-0.739	-0.596	-0.662	-0.519	-0.557	-0.456
<i>t-statistic</i>	-6.334	-6.226	-5.096	-5.610	-4.449	-4.761	-3.893
Panel C: Future Investment							
	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII
<i>Lender Shock</i>	-0.407*** (0.107)	-0.465*** (0.105)	-0.351*** (0.107)	-0.502*** (0.104)	-0.368*** (0.100)	-0.404*** (0.099)	-0.296*** (0.101)
<i>Macro Q</i>	0.126*** (0.004)	0.123*** (0.005)	0.118*** (0.005)	0.124*** (0.005)	0.124*** (0.004)	0.127*** (0.004)	0.124*** (0.005)
<i>Cash Flow</i>	0.282*** (0.080)	0.256*** (0.077)	0.307*** (0.086)	0.288*** (0.082)	0.261*** (0.075)	0.203** (0.090)	0.318*** (0.079)
<i>Firm Size</i>	-0.235*** (0.024)	-0.219*** (0.023)	-0.187*** (0.023)	-0.196*** (0.023)	-0.154*** (0.023)	-0.133*** (0.023)	-0.159*** (0.024)
<i>Altman's Z-score</i>	1.475*** (0.113)	1.489*** (0.104)	1.670*** (0.103)	1.535*** (0.107)	1.577*** (0.112)	1.469*** (0.121)	1.364*** (0.191)
<i>Intercept</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Obs.</i>	23806	23426	23012	22407	21929	21404	20892
<i>Adj.R-square</i>	0.252	0.247	0.245	0.243	0.244	0.245	0.245

Table IV: Investment Regressions with Lender Characteristics

This table presents results of cross-sectional investment regressions including controls for lender characteristics. The sample consists of firm-quarter observations of nonfinancial firms between 1987 and 2016 in the merged Compustat-Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock in the past 90 days when the loan is initiated. All the flow (stock) variables are measured during the fiscal quarter (end of the previous fiscal quarter) except *Lender Capitalization*, *Tier 1 Capital Ratio*, and *Book Assets*, which are measured as changes. The dependent variable is investment in the quarter following the loan. Model VI-VIII measure investment from two to four quarters following the loan. Investment measures are multiplied by 100 for readability. Industry fixed effects based on 3-digit Standard Industrial Classification (SIC) codes are included in all models. t-statistics robust to within-firm correlation and heteroskedasticity are reported in parentheses. Variable definitions are in Appendix A.

	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
<i>Lender Shock</i>	-0.407*** (0.117)	-0.398*** (0.128)	-0.422*** (0.117)	-0.391*** (0.111)	-0.377*** (0.129)	-0.269** (0.119)	-0.285** (0.122)	-0.067 (0.118)
<i>Macro Q</i>	0.253*** (0.014)	0.255*** (0.016)	0.254*** (0.014)	0.254*** (0.013)	0.238*** (0.016)	0.243*** (0.015)	0.227*** (0.015)	0.212*** (0.014)
<i>Cash Flow</i>	0.122 (0.098)	0.115 (0.104)	0.125 (0.098)	0.072 (0.083)	0.078 (0.109)	0.335*** (0.104)	0.447*** (0.109)	0.670*** (0.108)
<i>Firm Size</i>	-0.348*** (0.027)	-0.361*** (0.030)	-0.346*** (0.027)	-0.454*** (0.027)	-0.478*** (0.034)	-0.369*** (0.032)	-0.369*** (0.033)	-0.347*** (0.031)
<i>Altman's Z-score</i>	1.231*** (0.140)	1.195*** (0.144)	1.228*** (0.140)	1.181*** (0.118)	0.936*** (0.148)	0.968*** (0.142)	0.906*** (0.142)	0.931*** (0.138)
<i>Sq-Macro Q</i>	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
<i>Lag Cash Flow</i>	0.114 (0.126)	0.083 (0.135)	0.113 (0.125)	0.089 (0.097)	0.032 (0.133)	-0.038 (0.120)	-0.060 (0.137)	-0.138 (0.127)
Δ <i>Lender Capitalization</i>	0.050 (0.166)				0.179 (0.204)	0.403** (0.185)	0.358* (0.193)	0.278 (0.184)
Δ <i>Lender Tier 1 Capital Ratio</i>		-0.088 (0.089)			0.010 (0.093)	-0.074 (0.088)	-0.026 (0.088)	0.085 (0.085)
Δ <i>Lender Book Assets</i>			1.107*** (0.399)		0.967** (0.457)	0.272 (0.436)	0.657 (0.448)	0.631 (0.428)
<i>All-in-drawn Spread</i>				-0.588*** (0.048)	-0.564*** (0.061)	-0.525*** (0.056)	-0.536*** (0.059)	-0.504*** (0.056)
<i>Intercept</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Obs.</i>	16429	13603	16442	21801	12663	12512	12317	12126
<i>Adj.R-square</i>	0.271	0.276	0.272	0.269	0.283	0.285	0.276	0.274

Table V: Investment Regressions with Macroeconomic Characteristics

This table presents results of cross-sectional investment regressions including controls for macroeconomic characteristics. The sample consists of firm-quarter observations of nonfinancial firms between 1987 and 2016 in the merged Compustat-Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock in the past 90 days when the loan is initiated. All the flow (stock) variables are measured during the fiscal quarter (end of the previous fiscal quarter). The dependent variable is investment in the quarter following the loan. Model VI-VIII measure investment from two to four quarters following the loan. Investment measures are multiplied by 100 for readability. Industry fixed effects based on 3-digit Standard Industrial Classification (SIC) codes are included in all models. *t*-statistics robust to within-firm correlation and heteroskedasticity are reported in parentheses. Variable definitions are in Appendix A.

	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
<i>Lender Shock</i>	-0.338*** (0.111)	-0.324*** (0.111)	-0.369*** (0.111)	-0.844*** (0.122)	-0.548*** (0.123)	-0.510*** (0.120)	-0.401*** (0.118)	-0.414*** (0.117)
<i>Macro Q</i>	0.294*** (0.013)	0.261*** (0.012)	0.264*** (0.013)	0.260*** (0.013)	0.281*** (0.013)	0.286*** (0.012)	0.264*** (0.013)	0.262*** (0.012)
<i>Cash Flow</i>	0.086 (0.077)	0.109 (0.077)	0.100 (0.077)	0.101 (0.077)	0.083 (0.076)	0.347*** (0.081)	0.446*** (0.086)	0.688*** (0.091)
<i>Firm Size</i>	-0.217*** (0.027)	-0.299*** (0.024)	-0.326*** (0.024)	-0.320*** (0.024)	-0.224*** (0.027)	-0.135*** (0.026)	-0.178*** (0.025)	-0.148*** (0.025)
<i>Altman's Z-score</i>	1.361*** (0.112)	1.403*** (0.113)	1.419*** (0.114)	1.422*** (0.112)	1.333*** (0.112)	1.331*** (0.116)	1.295*** (0.111)	1.355*** (0.106)
<i>Sq_Macro Q</i>	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
<i>Lag Cash Flow</i>	0.107 (0.090)	0.115 (0.090)	0.105 (0.092)	0.114 (0.089)	0.104 (0.092)	-0.055 (0.101)	-0.010 (0.091)	-0.201** (0.097)
<i>Quarterly GDP growth</i>	-0.000*** (0.000)				-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)
<i>Baa - Aaa credit spreads</i>		-1.403*** (0.091)			-0.905*** (0.094)	-0.498*** (0.086)	-0.725*** (0.081)	-0.261*** (0.082)
<i>S&P 500 return</i>			1.602*** (0.366)		0.654* (0.362)	0.684** (0.347)	2.494*** (0.349)	0.127 (0.353)
<i>Aggregate defaults (indicator)</i>				-0.654*** (0.060)	-0.355*** (0.063)	-0.369*** (0.061)	-0.211*** (0.061)	-0.396*** (0.061)
<i>Intercept</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Obs.</i>	23796	23796	23498	23796	23498	23172	22840	22412
<i>Adj.R-square</i>	0.269	0.269	0.261	0.265	0.275	0.277	0.271	0.265

Table VI: Changes in Investment Regressions

This table presents results of cross-sectional regressions of changes in investment with respect to pre-loan investment level. The sample consists of firm-quarter observations of nonfinancial firms between 1987 and 2016 in the merged Compustat-Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock in the past 90 days when the loan is initiated. Models I-IV (Models V-VIII) present the results for an indicator for default shock (natural logarithm of count of default shocks) to measure lender shocks. All the flow (stock) variables are measured during the fiscal quarter (end of the previous fiscal quarter). Models I-IV (Models V-VIII) use the changes in average investment level in the first quarter and average of two, three and four quarters following the loan, respectively. Investment measures are multiplied by 100 for readability. Industry fixed effects based on 3-digit Standard Industrial Classification (SIC) codes are included in all models. *t*-statistics robust to within-firm correlation and heteroskedasticity are reported in parentheses. Variable definitions are in Appendix A.

	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
	Loan+1	Loan+2	Loan+3	Loan+4	Loan+1	Loan+2	Loan+3	Loan+4
<i>Lender Shock</i>	-0.225** (0.098)	-0.199** (0.087)	-0.195** (0.079)	-0.166** (0.077)				
<i>Shock Count</i>					-0.238** (0.098)	-0.218** (0.087)	-0.214*** (0.080)	-0.197** (0.078)
<i>Macro Q</i>	0.192*** (0.011)	0.203*** (0.010)	0.203*** (0.010)	0.201*** (0.010)	0.192*** (0.011)	0.203*** (0.010)	0.203*** (0.010)	0.201*** (0.010)
<i>Cash Flow</i>	0.108 (0.083)	0.224*** (0.072)	0.269*** (0.073)	0.373*** (0.067)	0.108 (0.083)	0.224*** (0.072)	0.269*** (0.073)	0.373*** (0.067)
<i>Firm Size</i>	-0.190*** (0.020)	-0.170*** (0.018)	-0.172*** (0.018)	-0.174*** (0.017)	-0.190*** (0.020)	-0.170*** (0.018)	-0.172*** (0.018)	-0.174*** (0.017)
<i>Altman's Z-score</i>	1.231*** (0.102)	1.230*** (0.094)	1.215*** (0.091)	1.224*** (0.089)	1.231*** (0.102)	1.230*** (0.094)	1.215*** (0.091)	1.223*** (0.089)
<i>Sq_Macro Q</i>	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)
<i>Lag Cash Flow</i>	0.226** (0.091)	0.155** (0.077)	0.123* (0.074)	0.063 (0.069)	0.226** (0.091)	0.156** (0.077)	0.123* (0.074)	0.064 (0.069)
<i>Lag Investment</i>	-0.627*** (0.009)	-0.681*** (0.008)	-0.712*** (0.007)	-0.738*** (0.007)	-0.626*** (0.009)	-0.681*** (0.008)	-0.712*** (0.007)	-0.738*** (0.007)
<i>Intercept</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Obs.</i>	22596	22269	21901	21454	22596	22269	21901	21454
<i>Adj.R-square</i>	0.353	0.471	0.532	0.577	0.353	0.471	0.533	0.577

Table VII: Panel Investment Regressions

This table presents results of panel data investment regressions. The sample consists of firm-quarter observations of nonfinancial firms between 1987 and 2016 for the duration of each loan in the merged Compustat-Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock in the past 90 days when the loan is initiated. All the flow (stock) variables are measured during the fiscal quarter (end of the previous fiscal quarter). In Panel A, the dependent variable is investment in quarter $t+1$. In Panel B Models I-IV (V-VIII) the dependent variable is changes in average investment level in quarter $t+1$ and average of $t+2$ and $t+3$ quarters, respectively. In both Panels, Models I-VI (Models VII-IX) present the results for an indicator for default shock (natural logarithm of count of default shocks) to measure lender shocks. In Panel C, the control variables are measured with respect to whether they are a flow or stock measure with respect to investment, except the shock variable that is measured 90 days prior to the loan issuance. Investment measures are multiplied by 100 for readability. Industry fixed effects based on 3-digit Standard Industrial Classification (SIC) codes are included in the models. t -statistics robust to within-firm correlation and heteroskedasticity are reported in parentheses. Variable definitions are in Appendix A.

	Panel A: Investment Levels								
	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII	Model IX
<i>Lender Shock</i>	-0.287*** (0.096)	-0.394*** (0.090)	-0.265*** (0.090)	-0.251*** (0.089)	-0.285*** (0.088)	-0.247*** (0.089)	-0.144* (0.081)		
<i>Shock Count</i>								-0.253*** (0.097)	-0.132 (0.086)
<i>Macro Q</i>		0.140*** (0.003)	0.132*** (0.003)	0.128*** (0.003)	0.272*** (0.009)	0.127*** (0.003)	0.337*** (0.010)	0.270*** (0.009)	0.337*** (0.010)
<i>Cash Flow</i>		0.299*** (0.035)	0.396*** (0.035)	0.177*** (0.035)	0.103*** (0.035)	0.171*** (0.030)	0.101*** (0.031)	0.122*** (0.030)	0.101*** (0.031)
<i>Firm Size</i>			-0.214*** (0.016)	-0.147*** (0.016)	-0.163*** (0.016)	-0.141*** (0.016)	-0.676*** (0.041)	-0.156*** (0.016)	-0.676*** (0.041)
<i>Altman's Z-score</i>				1.493*** (0.077)	1.431*** (0.076)	1.484*** (0.083)	1.459*** (0.087)	1.437*** (0.082)	1.459*** (0.087)
<i>Sq_Macro Q</i>					-0.003*** (0.000)		-0.004*** (0.000)	-0.003*** (0.000)	-0.004*** (0.000)
<i>Lag Cash Flow</i>						0.036 (0.033)	-0.044 (0.034)	-0.013 (0.033)	-0.044 (0.034)
<i>Intercept</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Event-Quarter FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No
<i>Firm FE</i>	No	No	No	No	No	No	Yes	No	Yes
<i>Obs.</i>	202503	178731	178731	173258	173258	171038	170899	171038	170899
<i>Adj.R-square</i>	0.160	0.235	0.241	0.263	0.273	0.259	0.441	0.268	0.441

Table VII: Continued

	Panel B: Changes in Investment						Loan+1	Loan+2	Loan+3
	Loan+1	Loan+2	Loan+3	Loan+1	Loan+2	Loan+3			
<i>Lender Shock</i>	-11.783*	-5.992*	-4.065*	-12.694**	-6.441**	-4.352**			
	(6.283)	(3.184)	(2.160)	(6.297)	(3.190)	(2.164)			
<i>Shock Count</i>							-10.330	-5.372	-3.651
							(6.771)	(3.434)	(2.326)
<i>Macro Q</i>	20.025***	10.078***	6.758***	20.189***	10.161***	6.812***	20.183***	10.158***	6.810***
	(0.704)	(0.359)	(0.244)	(0.700)	(0.357)	(0.243)	(0.700)	(0.357)	(0.243)
<i>Cash Flow</i>	15.388***	8.596***	5.866***	15.252***	8.533***	5.829***	15.251***	8.533***	5.828***
	(2.959)	(1.488)	(1.021)	(2.964)	(1.492)	(1.024)	(2.964)	(1.492)	(1.024)
<i>Firm Size</i>	-10.381***	-5.532***	-3.862***	-8.276***	-4.481***	-3.163***	-8.307***	-4.496***	-3.173***
	(1.191)	(0.606)	(0.409)	(1.206)	(0.614)	(0.414)	(1.206)	(0.614)	(0.414)
<i>Altman's Z-score</i>	130.138***	65.333***	44.268***	128.688***	64.627***	43.804***	128.685***	64.625***	43.802***
	(6.932)	(3.351)	(2.109)	(6.896)	(3.335)	(2.101)	(6.895)	(3.335)	(2.101)
<i>Sq-Macro Q</i>	-0.221***	-0.112***	-0.075***	-0.223***	-0.113***	-0.076***	-0.223***	-0.112***	-0.076***
	(0.014)	(0.007)	(0.005)	(0.014)	(0.007)	(0.005)	(0.014)	(0.007)	(0.005)
<i>Lag Cash Flow</i>	4.455	2.178	1.565	4.573	2.253	1.618	4.571	2.252	1.617
	(3.253)	(1.670)	(1.131)	(3.262)	(1.676)	(1.135)	(3.262)	(1.676)	(1.135)
<i>Lag Investment</i>	33.518***	16.434***	10.718***	33.223***	16.288***	10.621***	33.226***	16.289***	10.622***
	(0.461)	(0.235)	(0.160)	(0.461)	(0.235)	(0.160)	(0.461)	(0.235)	(0.160)
<i>Intercept</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Event-Quarter FE</i>	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Obs.</i>	163576	160441	157243	163570	160435	157237	163570	160435	157237
<i>Adj.R-square</i>	0.352	0.348	0.345	0.353	0.349	0.345	0.353	0.349	0.345

Table VIII: Investment Regressions Excluding Defaulter's Industry and Region

This table presents results of cross-sectional investment regressions. The sample consists of firm-quarter observations of nonfinancial firms between 1987 and 2016 in the merged Compustat-Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock from a borrower not in the same industry nor geographical region in the past 90 days when the loan is initiated, where industry refers to 1-digit Standard Industrial Classification (SIC) code and region refers to the U.S. state where headquarters are located (foreign headquartered firms are grouped as one region). All the flow (stock) variables are measured during the fiscal quarter (end of the previous fiscal quarter). The dependent variable is investment in the quarter following the loan. Investment measures are multiplied by 100 for readability. Industry fixed effects based on 3-digit SIC codes are included in all models. *t*-statistics robust to within-firm correlation and heteroskedasticity are reported in parentheses. Variable definitions are in Appendix A.

	Model I	Model II	Model III	Model IV
	Loan+1	Loan+2	Loan+1	Loan+2
<i>Lender Shock</i>	-0.416*** (0.112)	-0.403*** (0.108)		
<i>Shock Count</i>			-0.512*** (0.117)	-0.496*** (0.115)
<i>Macro Q</i>	0.125*** (0.004)	0.126*** (0.004)	0.092*** (0.005)	0.095*** (0.005)
<i>Cash Flow</i>	0.210*** (0.073)	0.282*** (0.080)	0.050 (0.084)	0.118 (0.090)
<i>Firm Size</i>	-0.317*** (0.024)	-0.235*** (0.024)	-0.289*** (0.024)	-0.213*** (0.024)
<i>Altman's Z-score</i>	1.545*** (0.107)	1.475*** (0.113)	0.960*** (0.110)	0.996*** (0.108)
<i>Intercept</i>	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes
<i>Obs.</i>	24209	23901	23482	23081
<i>Adj.R-square</i>	0.256	0.252	0.197	0.202

Table IX: Cross-sectional Variation in Investment

This table presents variation in the results of cross-sectional investment regressions according to various cross-sectional loan, borrower, relationship, and lender characteristics. The sample consists of firm-quarter observations of nonfinancial firms between 1987 and 2016 in the merged Compustat-Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock in the past 90 days when the loan is initiated. All the flow (stock) variables are measured during the fiscal quarter (end of the previous fiscal quarter). The reported values correspond to investment response sensitivity with respect to *Lender Shock* variable interacted with two extreme subgroups of firms with respect to the characteristics variable. To form the subgroups, we create terciles based on the borrower and loan characteristics, when the measures are continuous or use the indicator for discrete measures. The high (low) group corresponds to loans/firms that comprise of the top (bottom) tercile. In Panel A, the loan characteristics variables include the all-in-drawn spread, financial covenant (indicator), secured (indicator), and maturity of the loan. In Panel B, agency problem measures include the Whited-Wu index, cash holdings, credit rating (indicator), and tangibility. In Panel C, the relationship measures include the strong relationship with lead lender (S.Relation Lead Lender), any relationship with lead lender (A.Relation Lead Lender), any relationship with syndicate participant (A. Relation Syn. Member), and distance between lead lender and borrower (Distance Lead Lender). In Panel D, the lender characteristics include the lender capitalization level and changes in *Lender Capitalization*, *Tier 1 Capital Ratio*, and *Book Assets*. In all the Panels, we include all control variables as in Table III (Model IV) interacted with the loan and borrower characteristics subgroups. Investment measures are multiplied by 100 for readability. Industry fixed effects based on 3-digit Standard Industrial Classification (SIC) codes are included in all models. *t*-statistics robust to within-firm correlation and heteroskedasticity are reported in parentheses. Variable definitions are in Appendix A.

Panel A: Loan Characteristics							
	All-in-drawn Spread		Financial Covenant		Secured		Maturity
<i>High</i>	-0.827*** (0.253)	<i>Yes</i>	-0.562*** (0.147)	<i>Yes</i>	-0.614*** (0.186)	<i>High</i>	-0.541 (0.403)
<i>Low</i>	-0.376*** (0.143)	<i>No</i>	-0.240 (0.166)	<i>No</i>	-0.399** (0.187)	<i>Low</i>	-0.407** (0.180)
<i>Test of diff</i>		<i>Test of diff</i>		<i>Test of diff</i>		<i>Test of diff</i>	
<i>F-stat</i>	2.380	<i>F-stat</i>	2.157	<i>F-stat</i>	0.698	<i>F-stat</i>	0.092
<i>p-value</i>	0.123	<i>p-value</i>	0.142	<i>p-value</i>	0.404	<i>p-value</i>	0.762
<i>Control</i>	Yes	<i>Control</i>	Yes	<i>Control</i>	Yes	<i>Control</i>	Yes
<i>Obs.</i>	22185	<i>Obs.</i>	24213	<i>Obs.</i>	24213	<i>Obs.</i>	24171
<i>Adj.R-sq.</i>	0.205	<i>Adj.R-sq.</i>	0.257	<i>Adj.R-sq.</i>	0.210	<i>Adj.R-sq.</i>	0.184
Panel B: Agency Problems							
	WW-Index		Cash Holding		Rated		Tangibility
<i>High</i>	-0.596** (0.251)	<i>Yes</i>	-0.577*** (0.218)	<i>Yes</i>	-0.459*** (0.127)	<i>High</i>	-0.043 (0.208)
<i>Low</i>	-0.534*** (0.136)	<i>No</i>	-0.656*** (0.174)	<i>No</i>	-0.379* (0.195)	<i>Low</i>	-0.384* (0.213)
<i>Test of diff</i>		<i>Test of diff</i>		<i>Test of diff</i>		<i>Test of diff</i>	
<i>F-stat</i>	0.047	<i>F-stat</i>	0.080	<i>F-stat</i>	0.119	<i>F-stat</i>	1.305
<i>p-value</i>	0.829	<i>p-value</i>	0.778	<i>p-value</i>	0.730	<i>p-value</i>	0.253
<i>Control</i>	Yes	<i>Control</i>	Yes	<i>Control</i>	Yes	<i>Control</i>	Yes
<i>Obs.</i>	23848	<i>Obs.</i>	22585	<i>Obs.</i>	24213	<i>Obs.</i>	23998
<i>Adj.R-sq.</i>	0.202	<i>Adj.R-sq.</i>	0.232	<i>Adj.R-sq.</i>	0.257	<i>Adj.R-sq.</i>	0.241

Table IX: Continued

Panel C: Relationship Characteristics							
	S.Relation Lead Lender		A.Relation Lead Lender		A.Relation Syn. Member		Distance Lead Lender
<i>Yes</i>	-0.652*** (0.193)	<i>Yes</i>	-0.793*** (0.228)	<i>Yes</i>	-0.749*** (0.191)	<i>Far</i>	-0.040 (0.484)
<i>No</i>	-0.327** (0.131)	<i>No</i>	-0.356*** (0.125)	<i>No</i>	-0.324** (0.130)	<i>Close</i>	-0.011 (0.355)
<i>Test of diff</i>		<i>Test of diff</i>		<i>Test of diff</i>		<i>Test of diff</i>	
<i>F-stat</i>	2.036	<i>F-stat</i>	2.909	<i>F-stat</i>	3.620	<i>F-stat</i>	0.002
<i>p-value</i>	0.154	<i>p-value</i>	0.088	<i>p-value</i>	0.057	<i>p-value</i>	0.961
<i>Control</i>	Yes	<i>Control</i>	Yes	<i>Control</i>	Yes	<i>Control</i>	Yes
<i>Obs.</i>	24213	<i>Obs.</i>	24159	<i>Obs.</i>	24159	<i>Obs.</i>	4619
<i>Adj.R-sq.</i>	0.257	<i>Adj.R-sq.</i>	0.256	<i>Adj.R-sq.</i>	0.257	<i>Adj.R-sq.</i>	0.229
Panel D: Lender Characteristics							
	Lender Capitalization		Δ Lender Capitalization		Δ Lender Tier 1 Capital Ratio		Δ Lender Book Assets
<i>High</i>	-0.858*** (0.208)	<i>Yes</i>	-0.581*** (0.210)	<i>Yes</i>	-0.554*** (0.214)	<i>High</i>	-0.726*** (0.178)
<i>Low</i>	-0.253 (0.197)	<i>No</i>	-0.200 (0.185)	<i>No</i>	-0.662*** (0.209)	<i>Low</i>	-0.500** (0.210)
<i>Test of diff</i>		<i>Test of diff</i>		<i>Test of diff</i>		<i>Test of diff</i>	
<i>F-stat</i>	4.507	<i>F-stat</i>	1.849	<i>F-stat</i>	0.140	<i>F-stat</i>	0.697
<i>p-value</i>	0.034	<i>p-value</i>	0.174	<i>p-value</i>	0.709	<i>p-value</i>	0.404
<i>Control</i>	Yes	<i>Control</i>	Yes	<i>Control</i>	Yes	<i>Control</i>	Yes
<i>Obs.</i>	17780	<i>Obs.</i>	17768	<i>Obs.</i>	15107	<i>Obs.</i>	17781
<i>Adj.R-sq.</i>	0.209	<i>Adj.R-sq.</i>	0.216	<i>Adj.R-sq.</i>	0.219	<i>Adj.R-sq.</i>	0.212

Table X: Cross-sectional Variation in Investment According to Credit Market Characteristics

This table presents variation in the results of cross-sectional investment regressions according to cross-sectional credit market characteristics. The sample consists of firm-quarter observations of nonfinancial firms between 1997 (when the leveraged loan markets began) and 2016 in the merged Compustat-Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock in the past 90 days when the loan is initiated. All the flow (stock) variables are measured during the fiscal quarter (end of the previous fiscal quarter). The reported values correspond to investment response sensitivity with respect to *Lender Shock* variable interacted with two extreme subgroups of firms with respect to the characteristics variable. To form the subgroups, we use the indicator for discrete grouping measures. We define leveraged loans following Dealscan, as loans made by more than one lender to unrated borrowers or borrowers rated BB+ or lower, and priced at a spread over + 150 bps. Covenant-lite loans are identified those that are tagged by Dealscan as “Covenant-lite” in the market-segment file (Berlin et al. (2020)). Institutional investor participation indicator is defined as those loans who include at least one lender who is not a commercial bank (commercial banks have primary Standard Industrial Classification (SIC) codes between 6011-6082 or 6712). Crisis period is an indicator that takes the value on one for fiscal years 2001, 2002, 2007, 2008, and 2009, and zero otherwise. In all the Panels, we include all control variables as in Table III (Model IV) interacted with the credit market characteristics subgroups. Investment measures are multiplied by 100 for readability. Industry fixed effects based on 3-digit SIC codes are included in all models. *t*-statistics robust to within-firm correlation and heteroskedasticity are reported in parentheses. Variable definitions are in Appendix A.

	Leveraged Loans		Covenant Lite		Institutional Inv. Participation		Financial Crises
<i>Yes</i>	-0.464*** (0.175)	<i>Covenant-lite</i>	0.299 (1.138)	<i>Yes</i>	0.158 (0.318)	<i>Crisis period</i>	0.046 (0.161)
<i>No</i>	-0.302** (0.120)	<i>Non-covenant-lite</i>	-0.398*** (0.110)	<i>No</i>	-0.443*** (0.116)	<i>Non crisis period</i>	-0.568*** (0.157)
<i>Test of diff</i>		<i>Test of diff</i>		<i>Test of diff</i>		<i>Test of diff</i>	
<i>F-stat</i>	0.718	<i>F-stat</i>	0.370	<i>F-stat</i>	3.151	<i>F-stat</i>	6.290
<i>p-value</i>	0.397	<i>p-value</i>	0.543	<i>p-value</i>	0.076	<i>p-value</i>	0.012
<i>Control</i>	Yes	<i>Control</i>	Yes	<i>Control</i>	Yes	<i>Control</i>	Yes
<i>Obs.</i>	17050	<i>Obs.</i>	17466	<i>Obs.</i>	17466	<i>Obs.</i>	17466
<i>Adj.R-sq.</i>	0.274	<i>Adj.R-sq.</i>	0.274	<i>Adj.R-sq.</i>	0.274	<i>Adj.R-sq.</i>	0.274

Table XI: Investment Regressions - Shocks by Syndicate Role

This table presents results of cross-sectional investment regressions. The sample consists of firm-quarter observations of nonfinancial firms between 1987 and 2016 in the merged Compustat-Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock in the past 90 days. Participant Shock is an indicator variable that takes one when a syndicate participant in the lending syndicate experiences one or more default shock in the past 90 days. All the flow (stock) variables are measured during the fiscal quarter (end of the previous fiscal quarter). In Models I, II, and VII, the dependent variable is investment in the quarter following the loan. In Models III-VI, the dependent variable is investment in two to five quarters following the loan. In Models III-VI, the control variables are measured with respect to whether they are a flow or stock measure with respect to investment, except the shock variables that are measured 90 days prior to the loan. Investment measures are multiplied by 100 for readability. Industry fixed effects based on 3-digit Standard Industrial Classification (SIC) codes are included in all models. *t*-statistics robust to within-firm correlation and heteroskedasticity are reported in parentheses. Variable definitions are in Appendix A.

	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII
	Loan+1	Loan+1	Loan+2	Loan+3	Loan+4	Loan+5	Loan+1
<i>Lead Shock</i>	-0.384*** (0.109)	-0.360*** (0.107)	-0.438*** (0.108)	-0.495*** (0.099)	-0.337*** (0.102)	-0.452*** (0.101)	
<i>Participant Shock</i>	-0.283** (0.124)	-0.278** (0.124)	-0.260** (0.121)	-0.185 (0.121)	-0.318*** (0.117)	-0.413*** (0.112)	
<i>Lead Shock Count</i>							-0.356*** (0.108)
<i>Participant Shock Count</i>							-0.280** (0.118)
<i>Macro Q</i>	0.112*** (0.006)	0.248*** (0.015)	0.113*** (0.006)	0.109*** (0.006)	0.110*** (0.006)	0.116*** (0.006)	0.248*** (0.015)
<i>Cash Flow</i>	0.536*** (0.113)	0.119 (0.120)	0.434*** (0.128)	0.345*** (0.106)	0.335*** (0.125)	0.338*** (0.110)	0.120 (0.120)
<i>Firm Size</i>	-0.470*** (0.031)	-0.416*** (0.032)	-0.327*** (0.032)	-0.318*** (0.030)	-0.286*** (0.029)	-0.249*** (0.029)	-0.417*** (0.032)
<i>Sq-Macro Q</i>		1.276*** (0.128)					1.274*** (0.128)
<i>Altman's Z-score</i>		-0.003*** (0.000)	1.258*** (0.141)	0.940*** (0.153)	1.232*** (0.141)	1.427*** (0.121)	-0.003*** (0.000)
<i>Lag Cash Flow</i>		0.384*** (0.144)					0.384*** (0.144)
<i>Intercept</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Obs.</i>	19209	18116	18123	17884	17609	17154	18116
<i>Adj.R-square</i>	0.253	0.278	0.261	0.257	0.261	0.255	0.278

Table XII: Future Covenant Violation

This table presents results of a borrowers' probability of future covenant violation (Panel A) and the time to covenant violation (Panel B) using panel data. The sample consists of firm-quarter observations of nonfinancial firms between 1994 and 2016 for the duration of each loan in the merged Compustat-Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock in the past 90 days when the loan is initiated. All the variables are measured during the end of the previous fiscal quarter, except loan characteristics. In Panel A, the dependent variable is an indicator for whether the current ratio covenant (Models I and IV), the net-worth covenant (Models II and V), and any covenant (Models III and VI) are violated. In Panel B, the dependent variable is the time to violation for a current ratio covenant (Models I and IV), a net-worth covenant (Models II and V), and any covenant (Models III and VI). The current ratio and net-worth covenant based models are estimated at a quarterly frequency. Any covenant based models are estimated at an annual frequency. Industry fixed effects based on 3-digit Standard Industrial Classification (SIC) codes are included in all models. *t*-statistics (*z*-statistics) robust to within-firm correlation and heteroskedasticity are reported in parentheses in Panel A (Panel B). Variable definitions are in Appendix A.

Panel A: Probability of Future Covenant Violation - Logit Regression Model						
	Model I	Model II	Model III	Model IV	Model V	Model VI
	Current Ratio	Net Worth	Any Other	Current Ratio	Net Worth	Any Other
<i>Shock</i>	0.105** (2.24)	0.019 (0.63)	0.007 (0.61)			
<i>Shock Count</i>				-0.013 (-0.26)	-0.016 (-0.45)	0.020 (1.30)
<i>Amount</i>	-0.034 (-0.91)	-0.134*** (-4.44)	-0.059*** (-4.79)	-0.039 (-0.71)	-0.075** (-2.13)	-0.035** (-1.97)
<i>Firm Size</i>	-0.025*** (-2.98)	-0.040*** (-6.54)	-0.021*** (-6.64)	0.036** (2.07)	-0.084*** (-5.13)	-0.013** (-2.16)
<i>Investment Grade</i>	-0.114 (-1.42)	-0.009 (-0.35)	-0.023*** (-2.91)	0.034 (0.48)	-0.002 (-0.05)	-0.048*** (-2.72)
<i>Cash Flow</i>	-0.072*** (-3.82)	-0.075*** (-5.19)	-0.009** (-2.26)	-0.044*** (-3.03)	-0.039*** (-3.76)	-0.001 (-0.71)
<i>Leverage</i>	0.279*** (5.10)	0.564*** (10.09)	0.062*** (3.62)	0.072 (1.08)	0.552*** (8.91)	0.022 (1.14)
<i>Current Ratio</i>	-0.102*** (-11.40)	-0.019*** (-3.70)	-0.009*** (-6.15)	-0.089*** (-9.43)	-0.024*** (-4.96)	-0.003* (-1.93)
<i>Tangible Net Worth</i>	0.000* (1.78)	0.000** (2.33)	0.000** (2.21)	-0.000 (-0.18)	-0.000* (-1.91)	-0.000 (-0.93)
<i>MtB</i>	0.013 (1.64)	-0.009** (-2.20)	-0.007*** (-3.80)	-0.011 (-1.16)	-0.003 (-0.80)	-0.005** (-2.15)
Intercept	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	No	No	No
Firm FE	No	No	No	Yes	Yes	Yes
Quarter/Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	8901	17173	62835	8870	17101	62742
Adj. <i>R</i> -square	0.328	0.236	0.077	0.509	0.603	0.585

Table XII: Continued

Panel B: Time to Covenant Violation - Cox Regression Model						
	Model I	Model II	Model III	Model IV	Model V	Model VI
	Current Ratio	Net Worth	Any Other	Current Ratio	Net Worth	Any Other
<i>Shock</i>	0.313** (2.01)	0.142 (0.50)	0.305 (0.64)			
<i>Shock Count</i>				0.329** (2.06)	0.238 (0.87)	0.425 (1.08)
<i>Amount</i>	-0.156 (-1.05)	-1.029*** (-2.96)	-1.335** (-2.02)	-0.153 (-1.03)	-1.032*** (-2.96)	-1.326** (-2.01)
<i>Firm Size</i>	-0.130*** (-2.58)	-0.387*** (-7.39)	-0.280*** (-2.70)	-0.129** (-2.57)	-0.387*** (-7.40)	-0.280*** (-2.71)
<i>Investment Grade</i>	-0.532 (-1.03)	-0.289 (-0.90)	-1.701*** (-3.03)	-0.540 (-1.05)	-0.288 (-0.90)	-1.696*** (-3.01)
<i>Cash Flow</i>	-0.039 (-0.86)	-0.178*** (-3.38)	0.001 (0.01)	-0.038 (-0.84)	-0.178*** (-3.38)	0.003 (0.03)
<i>Leverage</i>	0.228 (1.03)	3.905*** (10.97)	0.713 (1.23)	0.222 (1.00)	3.908*** (10.97)	0.705 (1.21)
<i>Current Ratio</i>	-2.026*** (-18.85)	-0.352*** (-4.30)	-0.413*** (-3.33)	-2.028*** (-18.87)	-0.351*** (-4.30)	-0.411*** (-3.32)
<i>Tangible Net Worth</i>	0.000** (2.00)	0.000*** (3.52)	-0.000 (-0.12)	0.000** (2.05)	0.000*** (3.52)	-0.000 (-0.14)
<i>MtB</i>	0.004 (0.05)	-0.213** (-2.32)	-0.386** (-2.26)	0.003 (0.05)	-0.214** (-2.33)	-0.387** (-2.27)
Intercept	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes
Quarter/Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	8902	17175	14973	8902	17175	14973
Pseudo R-square	0.124	0.102	0.130	0.124	0.102	0.130

Table XIII: Investment Regressions Using Lender Shock in the Previous Year

This table presents results of cross-sectional investment regressions. The sample consists of firm-quarter observations of non-financial firms between 1987 and 2016 in the merged Compustat-Dealscan database. *Lender Shock* is an indicator variable that takes one when a lead lender experiences one or more default shock in the past 360 days when the loan is initiated. All the flow (stock) variables are measured during the fiscal quarter (end of the previous fiscal quarter). The dependent variable is investment in the quarter following the loan in Models I-IV. The dependent variable is investment in the second quarter following the loan in Models V-VIII. Investment measures are multiplied by 100 for readability. Industry fixed effects based on 3-digit Standard Industrial Classification (SIC) codes are included in all models. *t*-statistics robust to within-firm correlation and heteroskedasticity are reported in parentheses. Variable definitions are in Appendix A.

	Model I	Model II	Model III	Model IV	Model V	Model VI	Model VII	Model VIII
<i>Lender Shock</i>	-0.203*** (0.070)		-0.206** (0.088)	-0.148** (0.074)	-0.179** (0.070)		-0.153* (0.085)	-0.154** (0.073)
<i>Lender Shock Count</i>		-0.243*** (0.061)				-0.307*** (0.061)		
<i>Macro Q</i>	0.266*** (0.012)	0.266*** (0.013)	0.238*** (0.016)	0.281*** (0.013)	0.284*** (0.012)	0.278*** (0.012)	0.244*** (0.015)	0.286*** (0.013)
<i>Cash Flow</i>	0.075 (0.074)	0.091 (0.078)	0.042 (0.108)	0.085 (0.076)	0.141* (0.084)	0.164* (0.089)	0.335*** (0.100)	0.349*** (0.081)
<i>Firm Size</i>	-0.319*** (0.023)	-0.322*** (0.025)	-0.479*** (0.034)	-0.226*** (0.027)	-0.266*** (0.023)	-0.246*** (0.024)	-0.369*** (0.032)	-0.137*** (0.026)
<i>Altman's Z-score</i>	1.479*** (0.108)	1.450*** (0.116)	0.895*** (0.148)	1.333*** (0.112)	1.312*** (0.116)	1.433*** (0.118)	0.965*** (0.140)	1.332*** (0.116)
<i>Sq_Macro Q</i>	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)	-0.003*** (0.000)
<i>Lag Cash Flow</i>	0.121 (0.085)	0.125 (0.091)	0.091 (0.127)	0.103 (0.092)	0.062 (0.078)	0.040 (0.083)	-0.022 (0.113)	-0.056 (0.101)
<i>Intercept</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Lender Controls</i>	No	No	Yes	No	No	No	Yes	No
<i>Macroeconomic Controls</i>	No	No	No	Yes	No	No	No	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Obs.</i>	24790	23180	12803	23498	24860	23247	12652	23172
<i>Adj. R-square</i>	0.257	0.263	0.282	0.274	0.259	0.266	0.285	0.277

Appendix A: Variable definitions

Variable	Description
<i>Aggregate defaults (indicator)</i>	An indicator with value one if the total number of defaults in all lender portfolios in the previous quarter is greater than the median of defaults
<i>All-in-drawn Spread</i>	Natural logarithm of loan spread over LIBOR plus the annual fee plus the upfront fee over the duration of the loan
<i>Altman's Z-score</i>	Sum of 3.3 times pre-tax income, sales, 1.4 times retained earnings, and 1.2 times net working capital all divided by total assets (Chava and Roberts (2008))
<i>Baa-Aaa Credit Spread</i>	Difference between Moody's Baa-rated and Aaa-rated corporate bond yields
<i>Cash Flow</i>	Ratio of income before extraordinary items plus depreciation and amortization to start-of-period net property, plant, and equipment
<i>Cash Holding</i>	Ratio of cash and cash equivalents to total assets
<i>Current Ratio</i>	Ratio of current assets to current liabilities
<i>Financial Covenant</i>	An indicator which takes the value of one if the loan has one or more financial covenants
<i>Firm Size</i>	Natural logarithm of net property, plant, and equipment
<i>Fixed Charge Coverage Ratio</i>	The sum of rolling four-quarter operating income before depreciation scaled by the sum of rolling four-quarter interest expenses and debt in current liabilities
<i>Investment</i>	Ratio of capital expenditures to start-of-period net property, plant, and equipment (Chava and Roberts (2008))
<i>Investment Grade</i>	An indicator which takes the value of one if the long-term S&P issuer rating of the firm is greater than or equal to BBB-
<i>Lead Shock</i>	One or more defaults on the lead lender portfolio in the previous quarter
<i>Lender Book Assets</i>	Natural logarithm of total assets of the lender (Schwert (2018))
<i>Lender Capitalization</i>	Ratio of market value of equity to book value of equity of lender (Schwert (2018))
<i>Lender Shock</i>	One or more defaults on lender portfolio in the previous year (Murfin (2012))
<i>Lender Tier 1 Capital Ratio</i>	Risk weighted Tier 1 capital ratio of the lender (Schwert (2018))
<i>Leverage</i>	Ratio of total debt to total assets
<i>Loan Amount</i>	Natural logarithm of loan amount
<i>Log (Participants)</i>	Natural logarithm of number of participants in the loan syndicate
<i>Log(Assets)</i>	Natural logarithm of total assets
<i>Macro Q</i>	Ratio of (market value of equity + book value of liabilities - net inventories) to net property, plant, and equipment
<i>Maturity</i>	Natural logarithm of loan maturity in months
<i>MtB</i>	Ratio of (market value of equity + book value of liabilities + preferred equity - deferred Taxes and investment tax credit) to total assets
<i>Net Worth</i>	Total assets minus total liabilities
<i>Participant Shock</i>	One or more defaults on the syndicate participant portfolio in the previous quarter
<i>Quarterly GDP Growth</i>	Real GDP growth rate measured two quarters prior to the new loan
<i>Rated</i>	An indicator that takes the value of one if a firm has a long-term S&P issuer rating
<i>ROA</i>	Ratio of operating income before depreciation to total assets
<i>S&P 500 Return</i>	Holding period quarterly return for S&P 500 index measured two quarters prior to the new loan
<i>Secured</i>	An indicator which takes the value of one if the loan is secured
<i>Shock Count</i>	Natural logarithm of one plus the number of defaults in the lender portfolio in the previous quarter or year
<i>Sq_Macro Q</i>	Square of Macro Q
<i>Tangibility</i>	Ratio of net property, plant, and equipment to total assets
<i>Tangible Net Worth</i>	Total assets minus total liabilities and intangible assets (Murfin (2012))
<i>WW-Index</i>	Whited and Wu (2006) index of financial constraints

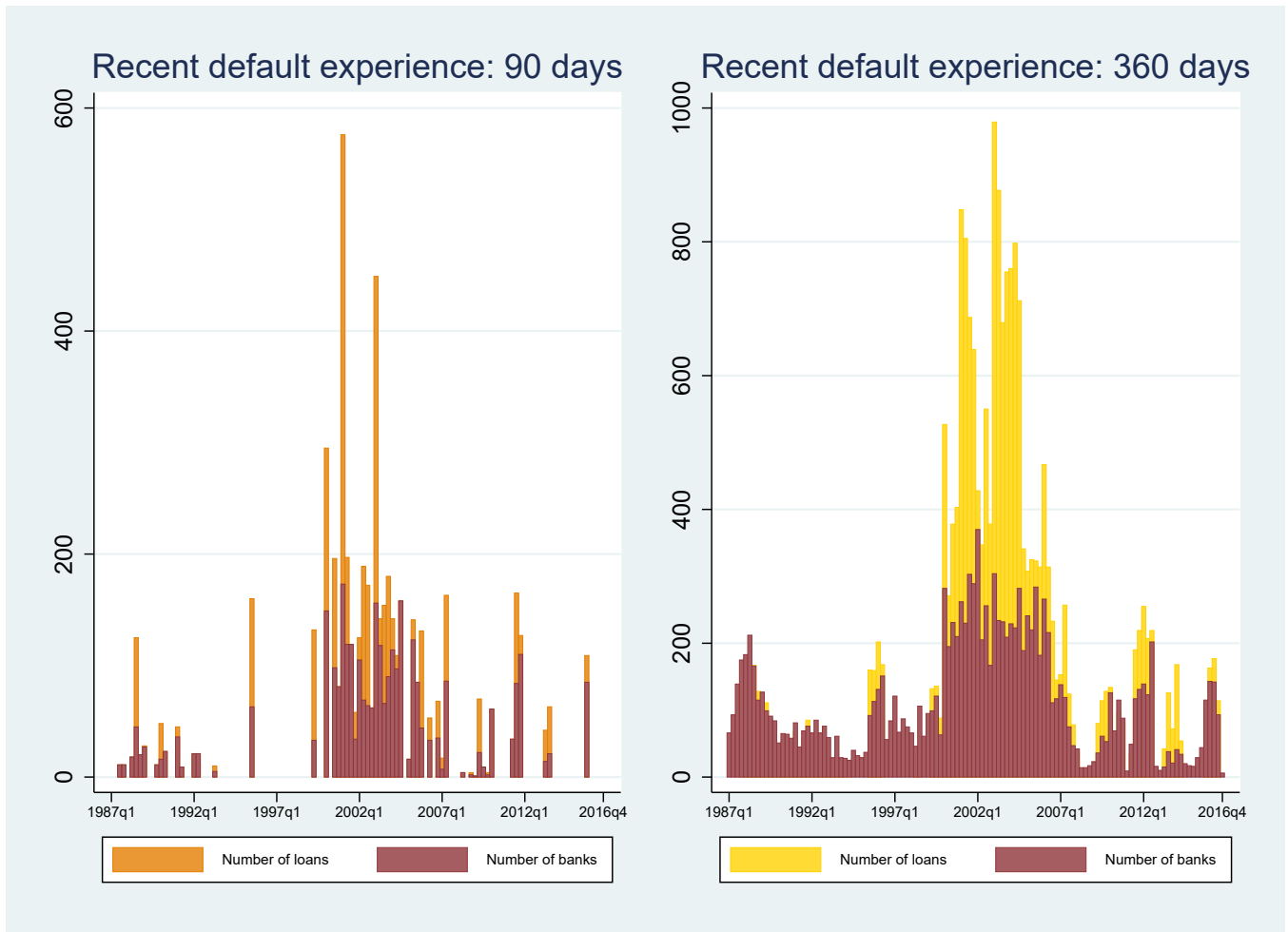


Figure 1. Distribution of Lender Default Shock

This figure presents the quarterly box-plot histogram of the distribution of default shocks and the affected lenders according to defaults in the 90 (left panel) and 360 (right panel) days prior to loan issuance during 1987 to 2016.

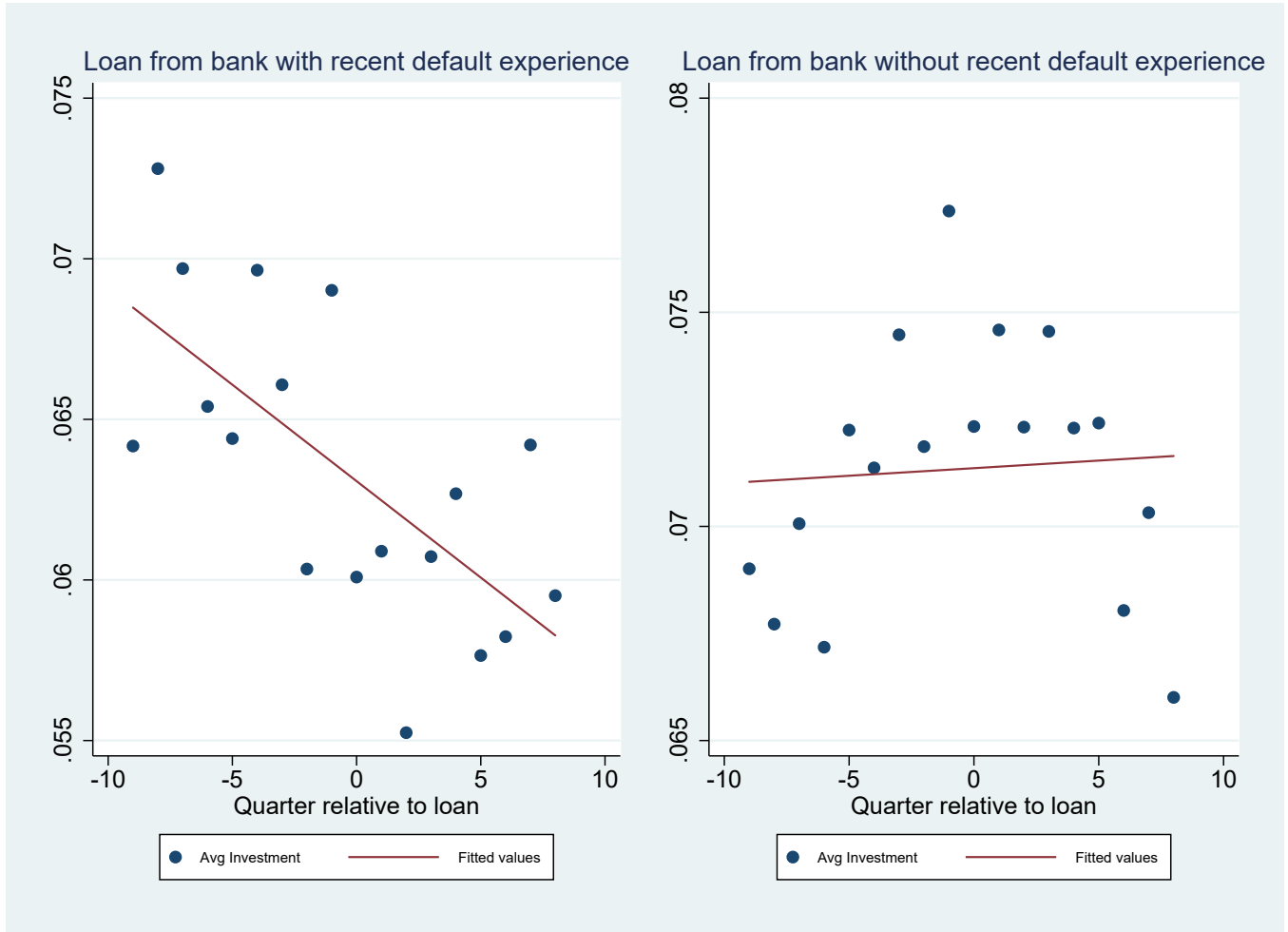


Figure 2. Average Investment According to Lender Default Experience

This figure presents the scatter plot of average investment in quarters around the loan borrowing. On the left (right) panel, we focus on eight quarters before and after borrowing a loan from a lender with (without) default experience in the prior 90 days during 1987 to 2016.